

APPENDIX TO THE REQUIREMENT GATHERING DOCUMENT (RGD v2.0)

GAP-07

Appendix-G

Data Dictionary / Data Catalogue

School ERP

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About This Appendix

This document is Appendix-G to the Requirement Gathering Document (RGD v2.0) for the School ERP System, addressing Gap-07: Data Dictionary / Data Catalogue. It defines every business entity, attribute, relationship, and lifecycle rule needed to support future Database Design, ER Diagramming, API Development, and Reporting.

This is a new, standalone document. RGD v2.0, Appendix-A through Appendix-F are not modified, opened for editing, or altered in producing this document. Every cross-reference (Related FRs, Related Business Rules) points back to those documents by ID only.

Scope discipline: 39 entities are fully specified because they trace to a Functional Requirement already approved in RGD v2.0/Appendix-E. Several concepts named in the request (Hostel, Visitor, Discipline, Complaint, Recruitment, Events) have no corresponding approved Functional Requirement — rather than inventing requirements that were never gathered, these are listed as clearly-flagged scope stubs in their own section, so nothing requested is silently omitted, but nothing is fabricated either.

Attribute catalogues below list each entity's core, business-meaningful fields (typically 4-9 per entity) sufficient to drive Database Design; exhaustive field-by-field enumeration of every possible technical column (e.g., internal soft-delete flags beyond what's shown) is deferred to the Database Design phase itself, consistent with this document's role as its primary input, not a replacement for it.

Naming Standards

Convention	Rule	Example
Entity Naming	PascalCase, singular noun	Student, Invoice, MarksRecord
Table Naming	snake_case, plural	students, invoices, marks_records
Column Naming	snake_case	admission_number, due_date
Primary Key Naming	<table_singular>_id	student_id, invoice_id
Foreign Key Naming	<referenced_table_singular>_id (or _ref_id for polymorphic)	class_id, owner_ref_id
Constraint Naming	uq_<table>_<column(s)> / ck_<table>_<rule>	uq_students_admission_number
Index Naming	idx_<table>_<column(s)>	idx_invoices_student_id

1. Entity Catalogue (Index)

Summary index of all entities. Full 28-field specifications and attribute catalogues follow in the Detailed Entity Specifications section.

Cross-Cutting / System

ID	Entity Name	Description	Data Category
ENT-SYS-001	User	System login/authentication account, one per human actor (staff, parent, or student) wh...	Master (System)
ENT-SYS-002	Role	Named permission set (e.g., Admin Staff, Finance Team, Academic Head) assigned to one o...	Master (System)
ENT-SYS-003	Guardian	Parent or legally responsible adult linked to one or more students, serving as the prim...	Master

ENT-SYS-004	AuditLog	Immutable record of every create/update/delete/approve/override action across the syste...	Transaction (Audit/System)
ENT-SYS-005	Configuration	System-wide key-value configuration store for all school-adjustable thresholds, rates, ...	Configuration
ENT-SYS-006	Document	Generic uploaded or system-generated document record (admission documents, ID cards, ce...	Transaction
ENT-SYS-007	ApprovalRequest	Generic approval/override workflow record used wherever a Business Rule requires elevat...	Transaction

Admission & Enrollment

ID	Entity Name	Description	Data Category
ENT-ADM-001	Application	Admission application submitted by a prospective student's parent/guardian.	Transaction
ENT-ADM-002	SeatAllocation	Class-wise seat capacity and live consumption tracking, including RTE-quota sub-allocat...	Master

Academic Master Data

ID	Entity Name	Description	Data Category
ENT-ACAD-001	AcademicSession	A single academic year/session (e.g., 2026-27), the top-level time boundary for nearly ...	Master
ENT-ACAD-002	Class	A grade level (e.g., Class 6), the primary academic grouping.	Master
ENT-ACAD-003	Section	A subdivision of a Class (e.g., 6-A), used for attendance, timetable, and capacity mana...	Master
ENT-ACAD-004	Subject	A curriculum subject (e.g., Mathematics), mapped to classes and used in Timetable and E...	Master
ENT-ACAD-005	GradingScheme	Board-aligned grading/marking scheme configuration (grade bands, GPA scale) used in res...	Master

Student Information Management

ID	Entity Name	Description	Data Category
ENT-SIS-001	Student	The core student master record: personal, medical, family, and academic identity data. ...	Master

Attendance Management

ID	Entity Name	Description	Data Category
ENT-ATT-001	AttendanceRecord	A single student's attendance state for one period on one date.	Transaction
ENT-ATT-002	StaffAttendanceRecord	A single staff member's attendance state for one day, reconciled against Leave.	Transaction

Timetable & Scheduling

ID	Entity Name	Description	Data Category
ENT-TT-001	TimetableEntry	A single scheduled period: class/section, subject, teacher, day, and period number.	Reference (versioned per term)

Examination & Result Processing

ID	Entity Name	Description	Data Category
ENT-EXM-001	Exam	A configured examination cycle for a class (dates, subjects, grading scheme).	Transaction
ENT-EXM-002	MarksRecord	A single student's marks for a single subject within an exam.	Transaction
ENT-EXM-003	ReportCard	A generated, and optionally published, report card document for a student for an exam.	Transaction
ENT-EXM-004	PromotionRecord	Record of a student's promotion from one class/session to the next.	Transaction

Fee Collection

ID	Entity Name	Description	Data Category
ENT-FEE-001	FeeHead	A category of fee (tuition, transport, exam fee) billed separately.	Master
ENT-FEE-002	FeeStructure	Configured fee amount per class, fee head, category, and academic session.	Master
ENT-FEE-003	Invoice	A generated fee invoice for a student, for one or more fee heads.	Transaction
ENT-FEE-004	Payment	A recorded payment transaction against an invoice, online or offline.	Transaction
ENT-FEE-005	ScholarshipWaiver	A fee waiver or scholarship applied to a student's fee head, including RTE-quota waivers.	Transaction

Library Management

ID	Entity Name	Description	Data Category
ENT-LIB-001	Book	A library catalog entry (title, author, classification, barcode).	Master
ENT-LIB-002	BookIssue	A single book issue/return transaction record.	Transaction

Transport Management

ID	Entity Name	Description	Data Category
ENT-TRN-001	Route	A configured bus route with defined stops.	Master
ENT-TRN-002	Vehicle	A transport vehicle with seating capacity and GPS device linkage.	Master
ENT-TRN-003	TransportAllocation	A student's confirmed allocation to a route/stop.	Transaction

HR & Payroll

ID	Entity Name	Description	Data Category
ENT-HR-001	Employee	The staff master record: identity, employment, and contract data.	Master

ENT-HR-002	Department	An organizational department (e.g., Academic, Administration, Finance).	Master
ENT-HR-003	Designation	A job title/designation (e.g., Teacher, Accountant, Librarian).	Master
ENT-HR-004	PayrollRun	A single staff member's payroll calculation for one pay period.	Transaction
ENT-HR-005	LeaveRequest	A staff leave application, approval, and balance-consumption record.	Transaction

Communication & Notifications

ID	Entity Name	Description	Data Category
ENT-COM-001	Circular	A homework, circular, or announcement posted to a targeted audience on the portal board.	Transaction
ENT-COM-002	NotificationLog	A dispatched (or attempted) notification via SMS, email, or app push.	Transaction

Out-of-Scope Stub Entities (named in request, no approved FR exists)

ID	Entity Name	Scope Status
ENT-OOS-001	Hostel	Not in RGD v2.0 Approved Scope
ENT-OOS-002	HostelRoom	Not in RGD v2.0 Approved Scope
ENT-OOS-003	Visitor	Not in RGD v2.0 Approved Scope
ENT-OOS-004	DisciplineRecord	Not in RGD v2.0 Approved Scope
ENT-OOS-005	Complaint	Not in RGD v2.0 Approved Scope
ENT-OOS-006	RecruitmentRequisition	Not in RGD v2.0 Approved Scope
ENT-OOS-007	Event	Not in RGD v2.0 Approved Scope

Detailed Entity Specifications — In Scope

Module: Cross-Cutting / System (7 Entities)**ENT-SYS-001 — User**

1. Entity Name	User
2. Description	System login/authentication account, one per human actor (staff, parent, or student) who accesses the ERP.
3. Purpose	Central identity record enabling authentication and role-based authorization across all modules.
4. Module	SYS
5. Owner Department (Business Owner)	IT Admin
6. Related Functional Requirements (Appx-E)	N/A (cross-cutting system entity)
7. Related Business Rules (Appx-C)	BR-HR-002
8. Primary Key	user_id
9. Foreign Keys	role_id → Role; owner_ref_id (polymorphic → Employee/Student/Guardian per owner_type)
10. Candidate Keys	username
11. Unique Constraints	username; email
12. Relationships	Many-to-One with Role; One-to-One with owning Employee/Student/Guardian record
13. Parent Entity	Role
14. Child Entities	AuditLog (via performed_by)
15. Data Category	Master (System)
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at account provisioning → Active → Locked (on failed attempts, NFR-PWD-003) → Deactivated (on exit, BR-HR-002) → Archived
18. Technical Owner	IT Admin
19. Retention Period	<i>Retained for audit purposes for 7 years post-deactivation (Client/Product Decision Required)</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Confidential
23. PII Classification	PII (username/email linked to identity)
24. Index Recommendations	Unique index on username, email; index on role_id
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Authentication API; SSO (future, per RGD v2.0 External Integrations)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: users (snake_case, plural) — see Naming Standards section

Attribute Catalogue — User

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
user_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	1001	Y	Y	N
username	Login identifier	VARCHAR	50	N	-	-	Unique, alphanumeric + dot/underscore	N/A	jdoe	Y	Y	N
password_hash	Hashed credential, never plaintext	VARCHAR	255	N	-	-	bcrypt/argon2 hash per NFR-PWD-001	NFR-PWD-001 (Appx-F)	(hash)	Y	N	Sensitive
role_id	FK to Role	BIGINT	-	N	-	-	Must exist in Role	N/A	3	Y	Y	N
owner_type	Polymorphic owner discriminator	VARCHAR	20	N	-	Employee/ Student/ Guardian	Enum	N/A	Employee	Y	Y	N
owner_ref_id	Polymorphic FK to owning record	BIGINT	-	N	-	-	Must exist in the table named by owner_type	N/A	4021	Y	Y	N
status	Account status	VARCHAR	20	N	'Active'	Active/Locked/ Deactivated	Enum	BR-HR-002	Active	Y	Y	N
failed_login_count	Consecutive failed login attempts	INT	-	N	0	0-5	Reset on success; lockout at 5	NFR-PWD-003 (Appx-F)	0	N	N	N
last_login_at	Timestamp of last successful login	DATETIME	-	Y	NULL	-	-	N/A	2026-07-20 08:01:00	N	N	N

ENT-SYS-002 — Role

1. Entity Name	Role
2. Description	Named permission set (e.g., Admin Staff, Finance Team, Academic Head) assigned to one or more Users.
3. Purpose	Implements role-based access control (RBAC) consistent with Appendix-E Field 38 permission matrices.
4. Module	SYS
5. Owner Department (Business Owner)	IT Admin
6. Related Functional Requirements (Appx-E)	N/A (cross-cutting system entity)
7. Related Business Rules (Appx-C)	BR-RPT-001, BR-FEE-002, BR-COM-002
8. Primary Key	role_id

9. Foreign Keys	None
10. Candidate Keys	role_name
11. Unique Constraints	role_name
12. Relationships	One-to-Many with User; Many-to-Many with Permission (via role_permission junction)
13. Parent Entity	None
14. Child Entities	User
15. Data Category	Master (System)
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at system setup → Active → Modified (permission changes logged) → rarely Deprecated
18. Technical Owner	IT Admin
19. Retention Period	Permanent — roles are structural, not transactional
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on role_name
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Authorization middleware on every API
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: roles (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Role

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
role_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	3	Y	Y	N
role_name	Display name of the role	VARCHAR	50	N	-	-	Unique	N/A	Finance Team	Y	Y	N
description	Role purpose description	VARCHAR	255	Y	NULL	-	-	N/A	Manages fee collection and financial reporting	N	N	N
is_system_role	Whether role is a protected system default	BOOLEAN	-	N	FALSE	TRUE/FALSE	-	N/A	TRUE	Y	N	N
permission_set	Serialized permission list (or	JSON	-	N	'{}'	-	Must map to defined	N/A	{"fee":	Y	N	N

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
	normalized via junction table)						Appendix-E Field 38 actions		["read","create"]			

ENT-SYS-003 — Guardian

1. Entity Name	Guardian
2. Description	Parent or legally responsible adult linked to one or more students, serving as the primary portal and communication contact.
3. Purpose	Enables guardian-student linkage (BR-SIS-006) for portal access and notification delivery.
4. Module	SYS
5. Owner Department (Business Owner)	Admin Staff
6. Related Functional Requirements (Appx-E)	FR-06
7. Related Business Rules (Appx-C)	BR-SIS-006
8. Primary Key	guardian_id
9. Foreign Keys	None (owns the M:N link)
10. Candidate Keys	mobile_number (within reason — see Uniqueness note)
11. Unique Constraints	None strictly enforced (siblings may share a guardian mobile number; email if provided should be unique)
12. Relationships	Many-to-Many with Student (via student_guardian_link junction table)
13. Parent Entity	None
14. Child Entities	Student (via junction)
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at admission (FR-01) → Active → Updated on contact change → Archived when all linked students leave
18. Technical Owner	IT Admin
19. Retention Period	<i>Retained per Data Retention policy (Client/Product Decision Required) after last linked student exits</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Confidential
23. PII Classification	PII
24. Index Recommendations	Index on mobile_number for lookup

25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Communication module (FR-37) recipient source; Parent Portal authentication
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: guardians (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Guardian

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
guardian_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	5001	Y	Y	N
full_name	Guardian's full name	VARCHAR	100	N	-	-	Non-empty	N/A	Ramesh Kumar	Y	N	PII
relationship	Relationship to student	VARCHAR	20	N	-	Father/Mother/Guardian/Other	Enum	N/A	Father	Y	N	PII
mobile_number	Primary contact number	VARCHAR	10	N	-	-	10-digit numeric	N/A	9876543210	Y	Y	PII
email	Contact email (optional)	VARCHAR	100	Y	NULL	-	Valid email format	N/A	ramesh.k@example.com	N	N	PII
is_primary_contact	Whether this is the primary notification recipient	BOOLEAN	-	N	FALSE	TRUE/FALSE	Exactly one primary per student	N/A	TRUE	Y	N	N

ENT-SYS-004 — AuditLog

1. Entity Name	AuditLog
2. Description	Immutable record of every create/update/delete/approve/override action across the system, per the standard audit template.
3. Purpose	Directly operationalizes Appendix-E Field 24 and Appendix-F NFR-AUDIT-001 across all modules.
4. Module	SYS
5. Owner Department (Business Owner)	IT Admin
6. Related Functional Requirements (Appx-E)	N/A (cross-cutting system entity)
7. Related Business Rules (Appx-C)	All immutability-related rules (BR-SIS-004, BR-ATT-002, BR-EXM-003, BR-FEE-001, BR-HR-002, BR-HR-007)
8. Primary Key	audit_log_id
9. Foreign Keys	performed_by → User
10. Candidate Keys	None
11. Unique Constraints	None

12. Relationships	Many-to-One with User (performed_by); polymorphic reference to any entity via entity_name + record_id
13. Parent Entity	None
14. Child Entities	None (terminal log)
15. Data Category	Transaction (Audit/System)
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created only — write-once, never updated or deleted under normal operation
18. Technical Owner	IT Admin
19. Retention Period	<i>Minimum 7 years for financial/statutory actions (Client/Product Decision Required to confirm exact period)</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Restricted
23. PII Classification	May contain PII within old_value/new_value snapshots
24. Index Recommendations	Composite index on (entity_name, record_id); index on performed_by; index on performed_at
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Consolidated audit-query API for IT Admin (NFR-LOG-010, Appx-F)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: audit_logs (snake_case, plural) — see Naming Standards section

Attribute Catalogue — AuditLog

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
audit_log_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	900123	Y	Y	N
entity_name	Name of the audited entity/table	VARCHAR	50	N	-	-	Must match a defined entity name	N/A	Invoice	Y	Y	N
record_id	Primary key value of the audited record	BIGINT	-	N	-	-	-	N/A	4501	Y	Y	N
action	Type of action performed	VARCHAR	20	N	-	Create/Update/Delete/Approve/Reject/Override	Enum	N/A	Update	Y	N	N
performed_by	FK to User who performed the action	BIGINT	-	N	-	-	Must exist in User	N/A	1001	Y	Y	N
performed_at	Timestamp of the action	DATETIME	-	N	CURRENT_TIMESTAMP	-	-	N/A	2026-07-23 10:15:00	Y	Y	N

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
					MP							
old_value	Prior field value(s), serialized	JSON	-	Y	NULL	-	-	N/A	{"status":"Draft"}	N	N	Sensitive (context-dependent)
new_value	New field value(s), serialized	JSON	-	Y	NULL	-	-	N/A	{"status":"Active"}	N	N	Sensitive (context-dependent)
ip_address	Origin IP address of the request	VARCHAR	45	Y	NULL	-	Valid IPv4/IPv6	N/A	103.25.12.4	N	N	N
reason	Reason for change, mandatory for corrections/overrides	VARCHAR	255	Y	NULL	-	Mandatory when action=Override	N/A	Re-evaluation per parent request	N	N	N

ENT-SYS-005 — Configuration

1. Entity Name	Configuration
2. Description	System-wide key-value configuration store for all school-adjustable thresholds, rates, and policy values.
3. Purpose	Operationalizes NFR-CFG-001 (Appx-F) and the 24 configurable items catalogued in Appendix-C v1.1 Section 3.5, avoiding hard-coded values.
4. Module	SYS
5. Owner Department (Business Owner)	IT Admin
6. Related Functional Requirements (Appx-E)	N/A (cross-cutting system entity)
7. Related Business Rules (Appx-C)	<i>All rules referencing Client/Product Decision Required in Appendix-C v1.1</i>
8. Primary Key	setting_id
9. Foreign Keys	None
10. Candidate Keys	setting_key
11. Unique Constraints	setting_key
12. Relationships	None (standalone reference entity, read by all modules)
13. Parent Entity	None
14. Child Entities	None
15. Data Category	Configuration
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at implementation → Modified by IT Admin as policy changes → versioned, not deleted
18. Technical Owner	IT Admin

19. Retention Period	Permanent, with full change history via AuditLog
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on setting_key
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Read by every module at runtime; admin UI for configuration management
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: configurations (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Configuration

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
setting_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	201	Y	Y	N
setting_key	Unique configuration key	VARCHAR	100	N	-	-	Unique, dot-notation namespaced	N/A	fee.late_fee_rate	Y	Y	N
setting_value	Current configured value	VARCHAR	500	N	-	-	Must match declared data_type	N/A	50.00	Y	N	N
data_type	Value's logical data type for parsing	VARCHAR	20	N	-	String/Number/Boolean/Date	Enum	N/A	Number	Y	N	N
module	Owning module for this setting	VARCHAR	30	N	-	-	Must match a defined module name	N/A	Fee Collection	Y	Y	N
is_editable	Whether IT Admin can change without code deployment	BOOLEAN	-	N	TRUE	TRUE/FALSE	-	N/A	TRUE	Y	N	N

ENT-SYS-006 — Document

1. Entity Name	Document
2. Description	Generic uploaded or system-generated document record (admission documents, ID cards, certificates, receipts).
3. Purpose	Single storage/reference pattern for all file-based artifacts across modules.
4. Module	SYS
5. Owner Department (Business)	Admin Staff

Owner)	
6. Related Functional Requirements (Appx-E)	FR-01, FR-09, FR-20, FR-23
7. Related Business Rules (Appx-C)	None Identified
8. Primary Key	document_id
9. Foreign Keys	owner_ref_id (polymorphic → Application/Student/Invoice per owner_type)
10. Candidate Keys	None
11. Unique Constraints	None
12. Relationships	Many-to-One (polymorphic) with owning entity
13. Parent Entity	Varies by owner_type
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created on upload/generation → Active → Archived per Data Retention policy
18. Technical Owner	IT Admin
19. Retention Period	Aligned to the owning entity's retention period
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Confidential
23. PII Classification	Context-dependent (e.g., admission documents are PII)
24. Index Recommendations	Index on (owner_type, owner_ref_id)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	<i>File storage service (local/cloud, Client/Product Decision Required); Certificate/Report Card generation (FR-09, FR-20)</i>
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: documents (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Document

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
document_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	70045	Y	Y	N
owner_type	Polymorphic owner	VARCHAR	30	N	-	Application/	Enum	N/A	Application	Y	Y	N

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
	discriminator					Student/ Invoice/ ReportCard						
owner_ref_id	Polymorphic FK to owning record	BIGINT	-	N	-	-	Must exist in the table named by owner_type	N/A	10023	Y	Y	N
document_type	Type of document	VARCHAR	50	N	-	Birth Certificate/ID Card/Report Card/Receipt/etc.	Enum	N/A	Birth Certificate	Y	N	N
file_path	Storage location reference	VARCHAR	500	N	-	-	Valid storage URI	N/A	/docs/2026/admission/10023.pdf	Y	N	N
uploaded_by	FK to User who uploaded/generated the document	BIGINT	-	N	-	-	Must exist in User	N/A	1001	Y	N	N
uploaded_at	Upload/generation timestamp	DATETIME	-	N	CURRENT_TIMESTAMP	-	-	N/A	2026-07-23 09:00:00	Y	N	N

ENT-SYS-007 — ApprovalRequest

1. Entity Name	ApprovalRequest
2. Description	Generic approval/override workflow record used wherever a Business Rule requires elevated authorization (leave, refund, re-evaluation, historical-record override).
3. Purpose	Provides a single, consistent workflow pattern for the many approval/override paths defined across Appendix-C.
4. Module	SYS
5. Owner Department (Business Owner)	Varies by module (see requested_by)
6. Related Functional Requirements (Appx-E)	FR-18, FR-25, FR-35, FR-08
7. Related Business Rules (Appx-C)	BR-ATT-003, BR-EXM-003, BR-HR-004, BR-SIS-004, BR-FEE-002
8. Primary Key	approval_request_id
9. Foreign Keys	requested_by → User; approver_id → User
10. Candidate Keys	None
11. Unique Constraints	None
12. Relationships	Many-to-One (polymorphic) with the entity/record being approved; Many-to-One with User (requester and approver)

13. Parent Entity	Varies by entity_name
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (Pending) → Approved/Rejected → Archived
18. Technical Owner	IT Admin
19. Retention Period	<i>Retained per AuditLog-equivalent retention policy (Client/Product Decision Required)</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Confidential
23. PII Classification	No (references records, not raw PII directly)
24. Index Recommendations	Index on (entity_name, record_id); index on status
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Notification module (FR-37) for approval-request and decision alerts
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: approval_requests (snake_case, plural) — see Naming Standards section

Attribute Catalogue — ApprovalRequest

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
approval_request_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	30012	Y	Y	N
entity_name	Entity type the approval applies to	VARCHAR	50	N	-	-	Must match a defined entity name	N/A	MarksRecord	Y	Y	N
record_id	Primary key of the record needing approval	BIGINT	-	N	-	-	-	N/A	55231	Y	Y	N
requested_by	FK to requesting User	BIGINT	-	N	-	-	Must exist in User	N/A	2044	Y	N	N
approver_id	FK to the User who decides	BIGINT	-	Y	NULL	-	Must hold the required approval role	N/A	1050	N	N	N
status	Current approval status	VARCHAR	20	N	'Pending'	Pending/ Approved/ Rejected	Enum	N/A	Pending	Y	Y	N
reason	Justification for the request	VARCHAR	255	N	-	-	Mandatory	N/A	Parent-requested re-evaluation	Y	N	N

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
decided_at	Timestamp of approval/rejection decision	DATETIME	-	Y	NULL	-	-	N/A	2026-07-23 14:00:00	N	N	N

Module: Admission & Enrollment (2 Entities)

ENT-ADM-001 — Application

1. Entity Name	Application
2. Description	Admission application submitted by a prospective student's parent/guardian.
3. Purpose	Captures and tracks an admission application from inquiry through confirmed enrollment.
4. Module	ADM
5. Owner Department (Business Owner)	Admin Staff
6. Related Functional Requirements (Appx-E)	FR-01, FR-02, FR-03, FR-05, FR-05a, FR-05b
7. Related Business Rules (Appx-C)	BR-ADM-005, BR-ADM-006
8. Primary Key	application_id
9. Foreign Keys	class_applied_id → Class
10. Candidate Keys	application_reference_no
11. Unique Constraints	application_reference_no
12. Relationships	Many-to-One with Class; One-to-One with Student (upon confirmation, FR-02)
13. Parent Entity	Class
14. Child Entities	Document (uploaded application documents)
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (Submitted) → Verified → Shortlisted/Waitlisted/Rejected → Admitted (locks and spawns Student record)
18. Technical Owner	IT Admin
19. Retention Period	<i>Rejected/withdrawn applications retained 2 years (Client/Product Decision Required); Admitted applications retained per Student retention policy</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Confidential
23. PII Classification	PII
24. Index Recommendations	Unique index on application_reference_no; index on status, class_applied_id
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	<i>Admission portal API; UIDAI verification (FR-05a, external, Client/Product Decision Required)</i>
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: applications (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Application

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
application_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	10023	Y	Y	N
application_reference_no	Applicant-facing tracking number	VARCHAR	20	N	-	-	Unique, system-generated	N/A	APP-2026-10023	Y	Y	N
applicant_name	Full name of the applicant	VARCHAR	100	N	-	-	Non-empty	N/A	Ananya Sharma	Y	N	PII
dob	Applicant date of birth	DATE	-	N	-	-	Past date; age-appropriate for class_applied_id	N/A	2019-04-12	Y	N	PII
class_applied_id	FK to Class applied for	BIGINT	-	N	-	-	Must exist in Class	N/A	6	Y	Y	N
aadhaar_number	Applicant/parent Aadhaar reference	VARCHAR	12	Y	NULL	-	12-digit numeric, checksum-valid	BR-ADM-006	XXXXXXXX1234	N	Y	PII (Sensitive)
category	Admission category	VARCHAR	20	N	'General'	General/RTE	Enum	BR-ADM-003, BR-ADM-004	RTE	Y	Y	N
status	Current application status	VARCHAR	20	N	'Submitted'	Submitted/ Verified/ Shortlisted/ Waitlisted/ Admitted/ Rejected	Enum, forward-only transitions	N/A	Shortlisted	Y	Y	N
submitted_at	Submission timestamp	DATETIME	-	N	CURRENT_TIMESTAMP	-	-	N/A	2026-04-01 10:00:00	Y	N	N

ENT-ADM-002 — SeatAllocation

1. Entity Name	SeatAllocation
2. Description	Class-wise seat capacity and live consumption tracking, including RTE-quota sub-allocation.
3. Purpose	Enforces BR-ADM-001 (capacity ceiling) and BR-ADM-003 (RTE ceiling) at the data layer.
4. Module	ADM
5. Owner Department (Business Owner)	Admin Staff
6. Related Functional Requirements (Appx-E)	FR-04

7. Related Business Rules (Appx-C)	BR-ADM-001, BR-ADM-003, BR-ADM-007, BR-ADM-008
8. Primary Key	seat_allocation_id
9. Foreign Keys	class_id → Class; academic_session_id → AcademicSession
10. Candidate Keys	(class_id, academic_session_id)
11. Unique Constraints	(class_id, academic_session_id)
12. Relationships	One-to-One with (Class, AcademicSession) pair; One-to-Many with Application via class_applied_id
13. Parent Entity	Class, AcademicSession
14. Child Entities	None
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at session setup → Updated on each confirmed admission → Closed at session end
18. Technical Owner	IT Admin
19. Retention Period	Permanent (historical capacity record)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique composite index on (class_id, academic_session_id)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Admission confirmation transaction (FR-02)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: seat_allocations (snake_case, plural) — see Naming Standards section

Attribute Catalogue — SeatAllocation

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
seat_allocation_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	501	Y	Y	N
class_id	FK to Class	BIGINT	-	N	-	-	Must exist in Class	N/A	6	Y	Y	N
academic_session_id	FK to AcademicSession	BIGINT	-	N	-	-	Must exist in AcademicSession	N/A	12	Y	Y	N
total_capacity	Configured total seats	INT	-	N	-	-	Positive integer	BR-ADM-001	40	Y	N	N

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
rte_quota_capacity	Configured RTE-quota seats	INT	-	N	-	-	≤25% of total_capacity	BR-ADM-003	10	Y	N	N
seats_filled	Current confirmed admissions	INT	-	N	0	-	≤ total_capacity	BR-ADM-001	32	Y	N	N
rte_seats_filled	Current confirmed RTE-quota admissions	INT	-	N	0	-	≤ rte_quota_capacity	BR-ADM-003	8	Y	N	N

Module: Academic Master Data (5 Entities)**ENT-ACAD-001 — AcademicSession**

1. Entity Name	AcademicSession
2. Description	A single academic year/session (e.g., 2026-27), the top-level time boundary for nearly all transactional data.
3. Purpose	Anchors every transactional entity to the correct academic year for reporting and historical retention.
4. Module	ACAD
5. Owner Department (Business Owner)	Academic Head
6. Related Functional Requirements (Appx-E)	N/A (implicit foundational master data across all FRs)
7. Related Business Rules (Appx-C)	BR-SIS-001
8. Primary Key	academic_session_id
9. Foreign Keys	None
10. Candidate Keys	session_name
11. Unique Constraints	session_name; (start_date, end_date) non-overlapping
12. Relationships	One-to-Many with nearly every transactional entity
13. Parent Entity	None
14. Child Entities	Application, SeatAllocation, AttendanceRecord, Exam, Invoice, PromotionRecord, etc.
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created ahead of the session → Active → Closed (triggers archival eligibility per NFR-ARC-001) → Archived
18. Technical Owner	IT Admin
19. Retention Period	Permanent (historical sessions never purged, only archived)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on session_name
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Referenced by nearly every module's API
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: academic_sessions (snake_case, plural) — see Naming Standards section

Attribute Catalogue — AcademicSession

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
academic_session_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	12	Y	Y	N
session_name	Display name of the session	VARCHAR	20	N	-	-	Unique, format YYYY-YY	N/A	2026-27	Y	Y	N
start_date	Session start date	DATE	-	N	-	-	Before end_date	N/A	2026-04-01	Y	N	N
end_date	Session end date	DATE	-	N	-	-	After start_date	N/A	2027-03-31	Y	N	N
status	Session lifecycle status	VARCHAR	20	N	'Planned'	Planned/Active/ Closed/Archived	Enum	BR-SIS-001	Active	Y	Y	N

ENT-ACAD-002 — Class

1. Entity Name	Class
2. Description	A grade level (e.g., Class 6), the primary academic grouping.
3. Purpose	Structural master data referenced by Admission, SIS, Timetable, Fee, and Examination.
4. Module	ACAD
5. Owner Department (Business Owner)	Academic Head
6. Related Functional Requirements (Appx-E)	FR-06, FR-14, FR-22
7. Related Business Rules (Appx-C)	None Identified
8. Primary Key	class_id
9. Foreign Keys	None
10. Candidate Keys	class_name
11. Unique Constraints	class_name
12. Relationships	One-to-Many with Section; One-to-Many with Application, Student (via Section), FeeStructure
13. Parent Entity	None
14. Child Entities	Section
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)

17. Lifecycle	Created at implementation → rarely modified → never deleted (only deactivated if curriculum changes)
18. Technical Owner	IT Admin
19. Retention Period	Permanent
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on class_name
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Referenced across Admission, SIS, Timetable, Fee, Examination APIs
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: classs (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Class

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
class_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	6	Y	Y	N
class_name	Display name	VARCHAR	20	N	-	-	Unique	N/A	Class 6	Y	Y	N
sequence_order	Ordering for promotion logic	INT	-	N	-	-	Unique, sequential	N/A	6	Y	N	N

ENT-ACAD-003 — Section

1. Entity Name	Section
2. Description	A subdivision of a Class (e.g., 6-A), used for attendance, timetable, and capacity management.
3. Purpose	Enables class-size management and homeroom-level operations.
4. Module	ACAD
5. Owner Department (Business Owner)	Academic Head
6. Related Functional Requirements (Appx-E)	FR-06, FR-14
7. Related Business Rules (Appx-C)	BR-SIS-005
8. Primary Key	section_id

9. Foreign Keys	class_id → Class
10. Candidate Keys	(class_id, section_name)
11. Unique Constraints	(class_id, section_name)
12. Relationships	Many-to-One with Class; One-to-Many with Student, TimetableEntry
13. Parent Entity	Class
14. Child Entities	Student, TimetableEntry
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at implementation → Modified (capacity changes) → rarely deactivated
18. Technical Owner	IT Admin
19. Retention Period	Permanent
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique composite index on (class_id, section_name)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Referenced across SIS, Attendance, Timetable, Examination APIs
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: sections (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Section

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
section_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	62	Y	Y	N
class_id	FK to Class	BIGINT	-	N	-	-	Must exist in Class	N/A	6	Y	Y	N
section_name	Section label	VARCHAR	10	N	-	-	Unique within class_id	N/A	A	Y	Y	N
capacity	Maximum students for this section	INT	-	N	-	-	Positive integer	BR-SIS-005	40	Y	N	N

ENT-ACAD-004 — Subject

1. Entity Name	Subject
2. Description	A curriculum subject (e.g., Mathematics), mapped to classes and used in Timetable and Examination.
3. Purpose	Structural master data underpinning timetable scheduling and marks entry.
4. Module	ACAD
5. Owner Department (Business Owner)	Academic Head
6. Related Functional Requirements (Appx-E)	FR-17, FR-14
7. Related Business Rules (Appx-C)	None Identified
8. Primary Key	subject_id
9. Foreign Keys	None (class mapping via ClassSubjectMap junction, described in Relationships)
10. Candidate Keys	subject_code
11. Unique Constraints	subject_code
12. Relationships	Many-to-Many with Class (via class_subject_map junction); One-to-Many with TimetableEntry, MarksRecord
13. Parent Entity	None
14. Child Entities	TimetableEntry, MarksRecord
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at curriculum setup → Modified (rare) → Deactivated if curriculum changes
18. Technical Owner	IT Admin
19. Retention Period	Permanent
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on subject_code
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Referenced across Timetable, Examination APIs
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: subjects (snake_case, plural) — see Naming Standards section

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
subject_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	14	Y	Y	N
subject_name	Display name	VARCHAR	50	N	-	-	Non-empty	N/A	Mathematics	Y	N	N
subject_code	Short unique code	VARCHAR	10	N	-	-	Unique	N/A	MATH	Y	Y	N

ENT-ACAD-005 — GradingScheme

1. Entity Name	GradingScheme
2. Description	Board-aligned grading/marking scheme configuration (grade bands, GPA scale) used in result calculation.
3. Purpose	Ensures results are computed per the school's confirmed board affiliation (BR-EXM-007).
4. Module	ACAD
5. Owner Department (Business Owner)	Academic Head
6. Related Functional Requirements (Appx-E)	FR-17, FR-19
7. Related Business Rules (Appx-C)	BR-EXM-007
8. Primary Key	grading_scheme_id
9. Foreign Keys	None
10. Candidate Keys	scheme_name
11. Unique Constraints	scheme_name
12. Relationships	One-to-Many with Exam
13. Parent Entity	None
14. Child Entities	Exam
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created per board affiliation → Modified only via a versioned update → historical versions retained for closed exams
18. Technical Owner	IT Admin
19. Retention Period	Permanent (historical schemes must remain available for closed-year report cards)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal

23. PII Classification	No
24. Index Recommendations	Unique index on scheme_name
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Grade/GPA calculation engine (FR-19)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: grading_schemes (snake_case, plural) — see Naming Standards section

Attribute Catalogue — GradingScheme

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
grading_scheme_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	2	Y	Y	N
scheme_name	Display name	VARCHAR	50	N	-	-	Unique	N/A	CBSE Standard Grading	Y	Y	N
board_type	Board this scheme aligns to	VARCHAR	20	N	-	CBSE/ICSE/State Board	Enum	BR-EXM-007	CBSE	Y	N	N
grade_band_json	Grade cut-off bands, serialized	JSON	-	N	-	-	Must be non-overlapping, ascending ranges	BR-EXM-007	{"A1": "91-100"}	Y	N	N

Module: Student Information Management (1 Entity)

ENT-SIS-001 — Student

1. Entity Name	Student
2. Description	The core student master record: personal, medical, family, and academic identity data. The single most-depended-upon entity in the system.
3. Purpose	Single system of record consumed by every other module (Appendix-D Critical Dependency List).
4. Module	SIS
5. Owner Department (Business Owner)	Admin Staff
6. Related Functional Requirements (Appx-E)	FR-06, FR-07, FR-08, FR-09
7. Related Business Rules (Appx-C)	BR-SIS-002, BR-SIS-003, BR-SIS-005, BR-SIS-006
8. Primary Key	student_id
9. Foreign Keys	section_id → Section; application_id → Application
10. Candidate Keys	admission_number; aadhaar_number
11. Unique Constraints	admission_number; aadhaar_number (where provided)
12. Relationships	One-to-One with Application; Many-to-One with Section; Many-to-Many with Guardian; One-to-Many with AttendanceRecord, MarksRecord, Invoice, BookIssue, TransportAllocation, PromotionRecord
13. Parent Entity	Section
14. Child Entities	AttendanceRecord, MarksRecord, Invoice, BookIssue, TransportAllocation, PromotionRecord, ReportCard, Document
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (Draft, at admission confirmation) → Active (BR-SIS-003) → Promoted (year-end) → Alumni/Exited → Archived
18. Technical Owner	IT Admin
19. Retention Period	<i>Retained per Data Retention policy (Client/Product Decision Required), typically several years post-exit for certificate re-issuance</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Restricted
23. PII Classification	PII (includes medical data — Sensitive PII)
24. Index Recommendations	Unique index on admission_number; unique index on aadhaar_number; index on section_id; search index on full_name (NFR-SRCH-001)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Consumed by Attendance, Examination, Fee, Transport, Library, Communication, Reports APIs

27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: students (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Student

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
student_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	4021	Y	Y	N
admission_number	Unique admission identifier	VARCHAR	20	N	-	-	Unique, system-generated at FR-02	BR-ADM-002 (Appx-C)	ADM-2026-04021	Y	Y	N
full_name	Student's full name	VARCHAR	100	N	-	-	Non-empty	N/A	Ananya Sharma	Y	N	PII
dob	Date of birth	DATE	-	N	-	-	Past date, age-appropriate for section_id's class	N/A	2019-04-12	Y	N	PII
aadhaar_number	Aadhaar reference (where provided)	VARCHAR	12	Y	NULL	-	12-digit numeric, checksum-valid, unique	BR-SIS-002	XXXXXXXX1234	N	Y	PII (Sensitive)
section_id	FK to current Section	BIGINT	-	N	-	-	Must exist in Section	BR-SIS-005	62	Y	Y	N
category	Admission category, carried from Application	VARCHAR	20	N	'General'	General/RTE	Enum	BR-ADM-003, BR-ADM-004	RTE	Y	Y	N
medical_info	Medical conditions/allergies	TEXT	-	Y	NULL	-	-	N/A	Peanut allergy	N	N	PII (Sensitive)
status	Student record status	VARCHAR	20	N	'Draft'	Draft/Active/Promoted/Exited/Archived	Enum, forward-only	BR-SIS-003	Active	Y	Y	N

Module: Attendance Management (2 Entities)**ENT-ATT-001 — AttendanceRecord**

1. Entity Name	AttendanceRecord
2. Description	A single student's attendance state for one period on one date.
3. Purpose	Feeds notifications, examination eligibility, and reporting.
4. Module	ATT
5. Owner Department (Business Owner)	Teacher / Academic Head
6. Related Functional Requirements (Appx-E)	FR-10, FR-11, FR-13
7. Related Business Rules (Appx-C)	BR-ATT-001, BR-ATT-002, BR-ATT-003, BR-ATT-007
8. Primary Key	attendance_record_id
9. Foreign Keys	student_id → Student; timetable_entry_id → TimetableEntry
10. Candidate Keys	(student_id, timetable_entry_id, attendance_date)
11. Unique Constraints	(student_id, timetable_entry_id, attendance_date)
12. Relationships	Many-to-One with Student; Many-to-One with TimetableEntry
13. Parent Entity	Student, TimetableEntry
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (marked) → Approved (locked, BR-ATT-002) → Correction (via ApprovalRequest if beyond window) → Archived
18. Technical Owner	IT Admin
19. Retention Period	10 million+ record scale per NFR-SCAL-005; archived per NFR-ARC-001 after academic session close
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No (references Student PII by ID only)
24. Index Recommendations	Unique composite index on (student_id, timetable_entry_id, attendance_date)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Notification trigger (FR-11); Attendance % reporting (FR-13); Payroll linkage (staff variant)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: attendance_records (snake_case, plural) — see Naming Standards section

Attribute Catalogue — AttendanceRecord

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
attendance_recor d_id	Surrogate primary key	BIGINT	-	N	auto- increment	-	System-generated	N/A	990001	Y	Y	N
student_id	FK to Student	BIGINT	-	N	-	-	Must exist in Student	N/A	4021	Y	Y	N
timetable_entry_i d	FK to TimetableEntry (identifies period/subject)	BIGINT	-	N	-	-	Must exist in TimetableEntry	N/A	3305	Y	Y	N
attendance_date	Calendar date of the record	DATE	-	N	-	-	Must be a scheduled school day	N/A	2026-07-23	Y	Y	N
state	Attendance state	VARCHAR	10	N	-	Present/Absent/ Late	Enum; single state per (student,timetable_entr y,date)	BR-ATT-001	Present	Y	N	N
marked_by	FK to User (Teacher) who marked it	BIGINT	-	N	-	-	Must exist in User	N/A	1050	Y	N	N
is_locked	Whether the record is approved/locked	BOOLEAN	-	N	FALSE	TRUE/FALSE	-	BR-ATT-002	TRUE	Y	N	N

ENT-ATT-002 — StaffAttendanceRecord

1. Entity Name	StaffAttendanceRecord
2. Description	A single staff member's attendance state for one day, reconciled against Leave.
3. Purpose	Feeds payroll calculation (BR-HR-001).
4. Module	ATT
5. Owner Department (Business Owner)	HR
6. Related Functional Requirements (Appx-E)	FR-12
7. Related Business Rules (Appx-C)	BR-ATT-005, BR-HR-001
8. Primary Key	staff_attendance_id
9. Foreign Keys	employee_id → Employee
10. Candidate Keys	(employee_id, attendance_date)
11. Unique Constraints	(employee_id, attendance_date)

12. Relationships	Many-to-One with Employee
13. Parent Entity	Employee
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (captured) → Reconciled (On Leave / Unauthorized) → Closed (month-end, gates PayrollRun) → Archived
18. Technical Owner	IT Admin
19. Retention Period	<i>Aligned with PayrollRun retention (statutory, Client/Product Decision Required)</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique composite index on (employee_id, attendance_date)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Payroll processing (FR-34)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: staff_attendance_records (snake_case, plural) — see Naming Standards section

Attribute Catalogue — StaffAttendanceRecord

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
staff_attendance_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	880001	Y	Y	N
employee_id	FK to Employee	BIGINT	-	N	-	-	Must exist in Employee	N/A	2044	Y	Y	N
attendance_date	Calendar date	DATE	-	N	-	-	-	N/A	2026-07-23	Y	Y	N
state	Reconciled attendance state	VARCHAR	20	N	-	Present/On Leave/Unauthorized	Enum	BR-ATT-005	Present	Y	N	N
is_reconciled	Whether cross-checked against Leave	BOOLEAN	-	N	FALSE	TRUE/FALSE	-	BR-ATT-005	TRUE	Y	N	N

Module: Timetable & Scheduling (1 Entity)

ENT-TT-001 — TimetableEntry

1. Entity Name	TimetableEntry
2. Description	A single scheduled period: class/section, subject, teacher, day, and period number.
3. Purpose	Structural backbone for Attendance and Examination scheduling.
4. Module	TT
5. Owner Department (Business Owner)	Academic Head
6. Related Functional Requirements (Appx-E)	FR-14, FR-15, FR-16
7. Related Business Rules (Appx-C)	BR-TT-001, BR-TT-002, BR-TT-003, BR-TT-005, BR-TT-006
8. Primary Key	timetable_entry_id
9. Foreign Keys	section_id → Section; subject_id → Subject; employee_id → Employee (teacher)
10. Candidate Keys	(section_id, day_of_week, period_no)
11. Unique Constraints	(section_id, day_of_week, period_no); (employee_id, day_of_week, period_no) — enforces BR-TT-001
12. Relationships	Many-to-One with Section, Subject, Employee
13. Parent Entity	Section, Subject, Employee
14. Child Entities	AttendanceRecord, MarksRecord (via Exam/Subject)
15. Data Category	Reference (versioned per term)
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (Draft) → Published (BR-TT-005, versioned) → Revised (new version) → Archived at term end
18. Technical Owner	IT Admin
19. Retention Period	Historical versions retained for the academic session; archived thereafter
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique composite index on (section_id, day_of_week, period_no); unique composite index on (employee_id, day_of_week, period_no)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Attendance capture (FR-10); Substitution management (FR-16)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: timetable_entries (snake_case, plural) — see Naming Standards section

Attribute Catalogue — TimetableEntry

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
timetable_entry_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	3305	Y	Y	N
section_id	FK to Section	BIGINT	-	N	-	-	Must exist in Section	N/A	62	Y	Y	N
subject_id	FK to Subject	BIGINT	-	N	-	-	Must exist in Subject	N/A	14	Y	Y	N
employee_id	FK to Employee (teacher)	BIGINT	-	N	-	-	Must exist in Employee; no double-booking	BR-TT-001	2044	Y	Y	N
day_of_week	Scheduled day	VARCHAR	10	N	-	Mon-Sat	Enum	N/A	Wednesday	Y	Y	N
period_no	Period number in the day	INT	-	N	-	1-8 (Client/Product Decision Required for max periods/day)	Positive integer	N/A	3	Y	Y	N
room_id	Room/lab identifier (if tracked)	VARCHAR	20	Y	NULL	-	No double-booking	BR-TT-002	LAB-2	N	N	N
version_no	Publication version number	INT	-	N	1	-	Increments on each revision	BR-TT-005	2	Y	N	N
status	Entry status	VARCHAR	20	N	'Draft'	Draft/Published	Enum	BR-TT-005	Published	Y	N	N

Module: Examination & Result Processing (4 Entities)

ENT-EXM-001 — Exam

1. Entity Name	Exam
2. Description	A configured examination cycle for a class (dates, subjects, grading scheme).
3. Purpose	Structural basis for marks entry and result calculation.
4. Module	EXM
5. Owner Department (Business Owner)	Academic Head
6. Related Functional Requirements (Appx-E)	FR-17
7. Related Business Rules (Appx-C)	BR-EXM-005, BR-EXM-007
8. Primary Key	exam_id
9. Foreign Keys	academic_session_id → AcademicSession; grading_scheme_id → GradingScheme; class_id → Class
10. Candidate Keys	(class_id, exam_name, academic_session_id)
11. Unique Constraints	(class_id, exam_name, academic_session_id)
12. Relationships	Many-to-One with AcademicSession, GradingScheme, Class; One-to-Many with MarksRecord
13. Parent Entity	AcademicSession, Class
14. Child Entities	MarksRecord, ReportCard
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (configured) → Active (marks entry open) → Locked (all subjects locked) → Closed (results published) → Archived
18. Technical Owner	IT Admin
19. Retention Period	Permanent (historical exam record)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique composite index on (class_id, exam_name, academic_session_id)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Marks entry (FR-18); Report card generation (FR-20)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: exams (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Exam

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
exam_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	55	Y	Y	N
exam_name	Display name	VARCHAR	50	N	-	-	Non-empty	N/A	Term 2 Final Examination	Y	N	N
class_id	FK to Class	BIGINT	-	N	-	-	Must exist in Class	N/A	6	Y	Y	N
academic_session_id	FK to AcademicSession	BIGINT	-	N	-	-	Must exist in AcademicSession	N/A	12	Y	Y	N
grading_scheme_id	FK to GradingScheme	BIGINT	-	N	-	-	Must match confirmed board affiliation	BR-EXM-007	2	Y	N	N
exam_date	Primary exam date (or start date for multi-day)	DATE	-	N	-	-	Within academic_session_id bounds	N/A	2026-12-10	Y	N	N
status	Exam cycle status	VARCHAR	20	N	'Configure d'	Configured/ Active/Locked/ Closed	Enum	N/A	Active	Y	Y	N

ENT-EXM-002 — MarksRecord

1. Entity Name	MarksRecord
2. Description	A single student's marks for a single subject within an exam.
3. Purpose	Source data for grade/rank/GPA calculation.
4. Module	EXM
5. Owner Department (Business Owner)	Teacher / Academic Head
6. Related Functional Requirements (Appx-E)	FR-18, FR-19
7. Related Business Rules (Appx-C)	BR-EXM-002, BR-EXM-003, BR-EXM-006
8. Primary Key	marks_record_id
9. Foreign Keys	exam_id → Exam; student_id → Student; subject_id → Subject
10. Candidate Keys	(exam_id, student_id, subject_id)
11. Unique Constraints	(exam_id, student_id, subject_id)

12. Relationships	Many-to-One with Exam, Student, Subject
13. Parent Entity	Exam, Student, Subject
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (entered) → Flagged (if anomalous, BR-EXM-006) → Locked (BR-EXM-003) → Re-evaluated (via ApprovalRequest) → Archived
18. Technical Owner	IT Admin
19. Retention Period	5 million+ record scale per NFR-SCAL-006; archived per NFR-ARC-001
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Restricted
23. PII Classification	No (references Student PII by ID only)
24. Index Recommendations	Unique composite index on (exam_id, student_id, subject_id)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Grade calculation engine (FR-19); Report card generation (FR-20)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: marks_records (snake_case, plural) — see Naming Standards section

Attribute Catalogue — MarksRecord

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
marks_record_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	770001	Y	Y	N
exam_id	FK to Exam	BIGINT	-	N	-	-	Must exist in Exam	N/A	55	Y	Y	N
student_id	FK to Student	BIGINT	-	N	-	-	Must exist in Student	N/A	4021	Y	Y	N
subject_id	FK to Subject	BIGINT	-	N	-	-	Must exist in Subject	N/A	14	Y	Y	N
marks_obtained	Marks scored	DECIMAL	5,2	Y	NULL	-	$0 \leq \text{value} \leq \text{max_marks}$; NULL = absent	BR-EXM-002	87.50	N	N	N
max_marks	Maximum marks configured for this subject/exam	DECIMAL	5,2	N	-	-	Positive	N/A	100.00	Y	N	N
is_flagged	Whether flagged as a statistical anomaly	BOOLEAN	-	N	FALSE	TRUE/FALSE	-	BR-EXM-006	FALSE	Y	N	N
is_locked	Whether marks are locked	BOOLEAN	-	N	FALSE	TRUE/FALSE	-	BR-EXM-003	TRUE	Y	N	N

ENT-EXM-003 — ReportCard

1. Entity Name	ReportCard
2. Description	A generated, and optionally published, report card document for a student for an exam.
3. Purpose	Compiles calculated results into the parent/student-facing document.
4. Module	EXM
5. Owner Department (Business Owner)	Academic Head
6. Related Functional Requirements (Appx-E)	FR-20, FR-21
7. Related Business Rules (Appx-C)	BR-EXM-001
8. Primary Key	report_card_id
9. Foreign Keys	student_id → Student; exam_id → Exam
10. Candidate Keys	(student_id, exam_id)
11. Unique Constraints	(student_id, exam_id)
12. Relationships	Many-to-One with Student, Exam; One-to-One with Document (generated PDF)
13. Parent Entity	Student, Exam
14. Child Entities	Document
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (generated) → Published (BR-EXM-001) → Archived at session close
18. Technical Owner	IT Admin
19. Retention Period	Permanent (academic record)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Restricted
23. PII Classification	PII
24. Index Recommendations	Unique composite index on (student_id, exam_id)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Parent/Student Portal (FR-21)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: report_cards (snake_case, plural) — see Naming Standards section

Attribute Catalogue — ReportCard

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
report_card_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	660001	Y	Y	N
student_id	FK to Student	BIGINT	-	N	-	-	Must exist in Student	N/A	4021	Y	Y	N
exam_id	FK to Exam	BIGINT	-	N	-	-	Must exist in Exam	N/A	55	Y	Y	N
grade_summary	Grade per subject, serialized	JSON	-	N	-	-	Must reflect locked MarksRecord values	N/A	{"MATH": "A1"}	Y	N	N
gpa	Calculated GPA	DECIMAL	3,2	N	-	-	Within GradingScheme scale	BR-EXM-004 (Appx-C)	9.20	Y	N	N
class_rank	Calculated class rank	INT	-	Y	NULL	-	Positive integer	N/A	3	N	N	N
is_published	Whether visible on the portal	BOOLEAN	-	N	FALSE	TRUE/FALSE	-	BR-EXM-001	TRUE	Y	Y	N
published_at	Publication timestamp	DATETIME	-	Y	NULL	-	-	N/A	2026-12-20 18:00:00	N	N	N

ENT-EXM-004 — PromotionRecord

1. Entity Name	PromotionRecord
2. Description	Record of a student's promotion from one class/session to the next.
3. Purpose	Enforces and documents BR-SIS-001 (academic/fee closure precondition).
4. Module	EXM
5. Owner Department (Business Owner)	Academic Head
6. Related Functional Requirements (Appx-E)	FR-08
7. Related Business Rules (Appx-C)	BR-SIS-001
8. Primary Key	promotion_record_id
9. Foreign Keys	student_id → Student; from_session_id → AcademicSession; to_session_id → AcademicSession; from_class_id → Class; to_class_id → Class
10. Candidate Keys	(student_id, from_session_id)
11. Unique Constraints	(student_id, from_session_id)
12. Relationships	Many-to-One with Student; Many-to-One with AcademicSession (twice, from/to); Many-to-One with Class (twice, from/to)
13. Parent Entity	Student
14. Child Entities	None
15. Data Category	Transaction

16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at year-end rollover → Confirmed → Archived
18. Technical Owner	IT Admin
19. Retention Period	Permanent (historical academic record, part of FR-08)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No (references Student PII by ID only)
24. Index Recommendations	Unique composite index on (student_id, from_session_id)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Academic year rollover process
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: promotion_records (snake_case, plural) — see Naming Standards section

Attribute Catalogue — PromotionRecord

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
promotion_record_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	44001	Y	Y	N
student_id	FK to Student	BIGINT	-	N	-	-	Must exist in Student	N/A	4021	Y	Y	N
from_session_id	FK to prior AcademicSession	BIGINT	-	N	-	-	Must exist in AcademicSession	N/A	12	Y	Y	N
to_session_id	FK to new AcademicSession	BIGINT	-	N	-	-	Must exist in AcademicSession	N/A	13	Y	Y	N
from_class_id	FK to prior Class	BIGINT	-	N	-	-	Must exist in Class	N/A	6	Y	N	N
to_class_id	FK to new Class	BIGINT	-	N	-	-	Must exist in Class; sequence_order = from + 1 (or repeat)	N/A	7	Y	N	N
academic_closure_confirmed	Whether academic records were closed before promotion	BOOLEAN	-	N	FALSE	TRUE/FALSE	Must be TRUE to promote	BR-SIS-001	TRUE	Y	N	N
fee_closure_confirmed	Whether fee records were closed before promotion	BOOLEAN	-	N	FALSE	TRUE/FALSE	Must be TRUE to promote	BR-SIS-001	TRUE	Y	N	N

Module: Fee Collection (5 Entities)

ENT-FEE-001 — FeeHead

1. Entity Name	FeeHead
2. Description	A category of fee (tuition, transport, exam fee) billed separately.
3. Purpose	Structural master data for fee configuration.
4. Module	FEE
5. Owner Department (Business Owner)	Finance Team
6. Related Functional Requirements (Appx-E)	FR-22
7. Related Business Rules (Appx-C)	None Identified
8. Primary Key	fee_head_id
9. Foreign Keys	None
10. Candidate Keys	fee_head_name
11. Unique Constraints	fee_head_name
12. Relationships	One-to-Many with FeeStructure
13. Parent Entity	None
14. Child Entities	FeeStructure
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at session setup → Modified (rare) → Deactivated if discontinued
18. Technical Owner	IT Admin
19. Retention Period	Permanent
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on fee_head_name
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Invoice generation (FR-23)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: fee_heads (snake_case, plural) — see Naming Standards section

Attribute Catalogue — FeeHead

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
fee_head_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	3	Y	Y	N
fee_head_name	Display name	VARCHAR	50	N	-	-	Unique	N/A	Tuition Fee	Y	Y	N
is_taxable	Whether GST applies	BOOLEAN	-	N	FALSE	TRUE/FALSE	-	BR-FEE-007	FALSE	Y	N	N
gst_rate	Applicable GST rate, if taxable	DECIMAL	5,2	Y	NULL	-	0-28 (Client/Product Decision Required to confirm applicable rate)	BR-FEE-007	18.00	N	N	N

ENT-FEE-002 — FeeStructure

1. Entity Name	FeeStructure
2. Description	Configured fee amount per class, fee head, category, and academic session.
3. Purpose	Billing foundation for automated invoice generation.
4. Module	FEE
5. Owner Department (Business Owner)	Finance Team
6. Related Functional Requirements (Appx-E)	FR-22
7. Related Business Rules (Appx-C)	None Identified
8. Primary Key	fee_structure_id
9. Foreign Keys	class_id → Class; fee_head_id → FeeHead; academic_session_id → AcademicSession
10. Candidate Keys	(class_id, fee_head_id, academic_session_id, category)
11. Unique Constraints	(class_id, fee_head_id, academic_session_id, category)
12. Relationships	Many-to-One with Class, FeeHead, AcademicSession
13. Parent Entity	Class, FeeHead, AcademicSession
14. Child Entities	Invoice (line items)
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)

17. Lifecycle	Created (versioned per session) → Active → Superseded (retained, not deleted, for historical invoice traceability)
18. Technical Owner	IT Admin
19. Retention Period	Permanent (needed to explain historical invoices)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique composite index on (class_id, fee_head_id, academic_session_id, category)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Invoice generation (FR-23)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: fee_structures (snake_case, plural) — see Naming Standards section

Attribute Catalogue — FeeStructure

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
fee_structure_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	900	Y	Y	N
class_id	FK to Class	BIGINT	-	N	-	-	Must exist in Class	N/A	6	Y	Y	N
fee_head_id	FK to FeeHead	BIGINT	-	N	-	-	Must exist in FeeHead	N/A	3	Y	Y	N
academic_session_id	FK to AcademicSession	BIGINT	-	N	-	-	Must exist in AcademicSession	N/A	12	Y	Y	N
category	Applicable student category	VARCHAR	20	N	'General'	General/RTE	Enum	BR-FEE-005	General	Y	Y	N
amount	Configured fee amount	DECIMAL	10,2	N	-	-	Positive number	N/A	5000.00	Y	N	N

ENT-FEE-003 — Invoice

1. Entity Name	Invoice
2. Description	A generated fee invoice for a student, for one or more fee heads.
3. Purpose	Core financial document driving payment collection.
4. Module	FEE
5. Owner Department (Business Owner)	Finance Team

6. Related Functional Requirements (Appx-E)	FR-23, FR-25, FR-26
7. Related Business Rules (Appx-C)	BR-FEE-001
8. Primary Key	invoice_id
9. Foreign Keys	student_id → Student; academic_session_id → AcademicSession
10. Candidate Keys	invoice_no
11. Unique Constraints	invoice_no
12. Relationships	Many-to-One with Student, AcademicSession; One-to-Many with Payment
13. Parent Entity	Student, AcademicSession
14. Child Entities	Payment
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (Generated) → Paid (locked, BR-FEE-001) / Overdue (Defaulter, BR-FEE-008) → Credited (via credit note, not edit) → Archived
18. Technical Owner	IT Admin
19. Retention Period	<i>500,000+ transaction scale per NFR-SCAL-007; statutory financial retention period (Client/Product Decision Required)</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Restricted
23. PII Classification	PII (linked to student financial data)
24. Index Recommendations	Unique index on invoice_no; index on student_id, status, due_date
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Payment Gateway (FR-24); Financial Reporting (FR-26)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: invoices (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Invoice

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
invoice_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	4501	Y	Y	N
invoice_no	Human-readable invoice number	VARCHAR	20	N	-	-	Unique, system-generated	N/A	INV-2026-04501	Y	Y	N

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
student_id	FK to Student	BIGINT	-	N	-	-	Must exist in Student	N/A	4021	Y	Y	N
academic_session_id	FK to AcademicSession	BIGINT	-	N	-	-	Must exist in AcademicSession	N/A	12	Y	Y	N
total_amount	Total invoice amount	DECIMAL	10,2	N	-	-	Positive, sum of line items minus waivers	N/A	5000.00	Y	N	N
due_date	Payment due date	DATE	-	N	-	-	-	N/A	2026-08-05	Y	Y	N
status	Invoice status	VARCHAR	20	N	'Unpaid'	Unpaid/Paid/ Partially Paid/Defaulter/C ancelled	Enum	BR-FEE-001, BR-FEE-008	Unpaid	Y	Y	N
is_locked	Whether locked from direct edits	BOOLEAN	-	N	FALSE	TRUE/FALSE	Becomes TRUE once a payment is recorded	BR-FEE-001	FALSE	Y	N	N

ENT-FEE-004 — Payment

1. Entity Name	Payment
2. Description	A recorded payment transaction against an invoice, online or offline.
3. Purpose	Records fee collection and gateway transaction linkage.
4. Module	FEE
5. Owner Department (Business Owner)	Finance Team
6. Related Functional Requirements (Appx-E)	FR-24
7. Related Business Rules (Appx-C)	BR-FEE-002, BR-FEE-006
8. Primary Key	payment_id
9. Foreign Keys	invoice_id → Invoice
10. Candidate Keys	gateway_transaction_ref
11. Unique Constraints	gateway_transaction_ref (where present, per BR-FEE-006)
12. Relationships	Many-to-One with Invoice
13. Parent Entity	Invoice
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)

17. Lifecycle	Created (recorded) → Reconciled → Refunded/Voided (Finance Team only, BR-FEE-002) → Archived
18. Technical Owner	IT Admin
19. Retention Period	<i>Statutory financial retention period (Client/Product Decision Required)</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Restricted
23. PII Classification	PII (financial data)
24. Index Recommendations	Unique index on gateway_transaction_ref; index on invoice_id
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	<i>Payment Gateway API (Client/Product Decision Required for vendor)</i>
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: payments (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Payment

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
payment_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	6001	Y	Y	N
invoice_id	FK to Invoice	BIGINT	-	N	-	-	Must exist in Invoice	N/A	4501	Y	Y	N
amount_paid	Amount of this payment	DECIMAL	10,2	N	-	-	Positive	N/A	5000.00	Y	N	N
payment_mode	Payment channel	VARCHAR	20	N	-	Online/Cash/ Cheque/Bank Transfer	Enum	N/A	Online	Y	N	N
gateway_transaction_ref	Gateway transaction reference (for online payments)	VARCHAR	100	Y	NULL	-	Unique when present	BR-FEE-006	TXN20260723001	N	Y	Sensitive
paid_at	Payment timestamp	DATETIME	-	N	CURRENT_TIMESTAMP	-	-	N/A	2026-07-23 11:00:00	Y	N	N
status	Payment status	VARCHAR	20	N	'Success'	Success/Failed/ Refunded/Voided	Enum	BR-FEE-002	Success	Y	N	N

ENT-FEE-005 — ScholarshipWaiver

1. Entity Name	ScholarshipWaiver
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2. Description	A fee waiver or scholarship applied to a student's fee head, including RTE-quota waivers.
3. Purpose	Tracks waived amounts separately for RTE reimbursement reporting (FR-26b).
4. Module	FEE
5. Owner Department (Business Owner)	Finance Team
6. Related Functional Requirements (Appx-E)	FR-26b
7. Related Business Rules (Appx-C)	BR-FEE-005
8. Primary Key	scholarship_waiver_id
9. Foreign Keys	student_id → Student; fee_head_id → FeeHead
10. Candidate Keys	None
11. Unique Constraints	None
12. Relationships	Many-to-One with Student, FeeHead
13. Parent Entity	Student, FeeHead
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (applied at invoicing) → Active for the session → Archived
18. Technical Owner	IT Admin
19. Retention Period	Aligned with Invoice retention (statutory)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Confidential
23. PII Classification	PII
24. Index Recommendations	Index on student_id
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Invoice generation (FR-23); RTE reimbursement report (FR-26b)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: scholarship_waivers (snake_case, plural) — see Naming Standards section

Attribute Catalogue — ScholarshipWaiver

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
scholarship_waiver_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	7001	Y	Y	N
student_id	FK to Student	BIGINT	-	N	-	-	Must exist in Student	N/A	4021	Y	Y	N
fee_head_id	FK to FeeHead	BIGINT	-	N	-	-	Must exist in FeeHead	N/A	3	Y	N	N
waiver_type	Type of waiver	VARCHAR	20	N	-	RTE/Merit/ Sibling/Staff- Ward	Enum	BR-FEE-005, BR-ADM-003	RTE	Y	N	N
waiver_amount	Amount waived	DECIMAL	10,2	N	-	-	Positive, ≤ FeeStructure amount for that head	BR-FEE-005	5000.00	Y	N	N

Module: Library Management (2 Entities)

ENT-LIB-001 — Book

1. Entity Name	Book
2. Description	A library catalog entry (title, author, classification, barcode).
3. Purpose	Structural master data for issue/return processing and self-service search.
4. Module	LIB
5. Owner Department (Business Owner)	Librarian
6. Related Functional Requirements (Appx-E)	FR-27, FR-29
7. Related Business Rules (Appx-C)	BR-LIB-004
8. Primary Key	book_id
9. Foreign Keys	None
10. Candidate Keys	barcode
11. Unique Constraints	barcode
12. Relationships	One-to-Many with BookIssue
13. Parent Entity	None
14. Child Entities	BookIssue
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (cataloged) → Active → Lost/Damaged (BR-LIB-003) → Withdrawn/Archived
18. Technical Owner	IT Admin
19. Retention Period	Permanent (catalog history)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on barcode; search index on title, author (NFR-SRCH-001)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Issue/return transaction (FR-28)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: books (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Book

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
book_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	1201	Y	Y	N
barcode	Unique physical identifier	VARCHAR	30	N	-	-	Unique	N/A	LIB-000001201	Y	Y	N
title	Book title	VARCHAR	200	N	-	-	Non-empty	N/A	Wings of Fire	Y	N	N
author	Author name(s)	VARCHAR	150	Y	NULL	-	-	N/A	A.P.J. Abdul Kalam	N	N	N
classification	Catalog classification	VARCHAR	30	N	'Circulating'	Circulating/Reference	Enum	BR-LIB-004	Circulating	Y	N	N
is_available	Current availability status	BOOLEAN	-	N	TRUE	TRUE/FALSE	-	N/A	TRUE	Y	N	N

ENT-LIB-002 — BookIssue

1. Entity Name	BookIssue
2. Description	A single book issue/return transaction record.
3. Purpose	Tracks circulation and overdue fine calculation.
4. Module	LIB
5. Owner Department (Business Owner)	Librarian
6. Related Functional Requirements (Appx-E)	FR-28
7. Related Business Rules (Appx-C)	BR-LIB-001, BR-LIB-002, BR-LIB-005
8. Primary Key	book_issue_id
9. Foreign Keys	book_id → Book; borrower_ref_id (polymorphic → Student/Employee per borrower_type)
10. Candidate Keys	None
11. Unique Constraints	None
12. Relationships	Many-to-One with Book; Many-to-One (polymorphic) with Student or Employee
13. Parent Entity	Book
14. Child Entities	None
15. Data Category	Transaction

16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (Issued) → Returned → Fine Applied (if overdue, BR-LIB-002) → Archived
18. Technical Owner	IT Admin
19. Retention Period	Permanent (borrowing history)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No (references borrower PII by ID only)
24. Index Recommendations	Index on book_id; index on (borrower_type, borrower_ref_id); index on due_date (for overdue scans)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Fee ledger (fine posting, cross-module per Appendix-D)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: book_issues (snake_case, plural) — see Naming Standards section

Attribute Catalogue — BookIssue

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
book_issue_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	330001	Y	Y	N
book_id	FK to Book	BIGINT	-	N	-	-	Must exist in Book; must be available	BR-LIB-004	1201	Y	Y	N
borrower_type	Polymorphic borrower discriminator	VARCHAR	20	N	-	Student/ Employee	Enum	N/A	Student	Y	Y	N
borrower_ref_id	Polymorphic FK to borrower record	BIGINT	-	N	-	-	Must exist in the table named by borrower_type; must be under max-books limit	BR-LIB-001	4021	Y	Y	N
issue_date	Date issued	DATE	-	N	CURRENT_DATE	-	-	N/A	2026-07-10	Y	N	N
due_date	Date due back	DATE	-	N	-	-	After issue_date	N/A	2026-07-24	Y	Y	N
return_date	Date actually returned	DATE	-	Y	NULL	-	-	N/A	2026-07-27	N	N	N
fine_amount	Calculated overdue fine	DECIMAL	8,2	N	0.00	-	≥0, per-day rate × days overdue	BR-LIB-002	150.00	Y	N	N

Module: Transport Management (3 Entities)**ENT-TRN-001 — Route**

1. Entity Name	Route
2. Description	A configured bus route with defined stops.
3. Purpose	Structural master data for transport allocation.
4. Module	TRN
5. Owner Department (Business Owner)	Transport Coordinator
6. Related Functional Requirements (Appx-E)	FR-30
7. Related Business Rules (Appx-C)	BR-TRN-001
8. Primary Key	route_id
9. Foreign Keys	None
10. Candidate Keys	route_name
11. Unique Constraints	route_name
12. Relationships	One-to-Many with TransportAllocation; One-to-One (or One-to-Many across sessions) with Vehicle
13. Parent Entity	None
14. Child Entities	TransportAllocation
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (configured) → Active → Modified (stops/capacity changes) → Deactivated
18. Technical Owner	IT Admin
19. Retention Period	Permanent
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on route_name
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Transport allocation (FR-30)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: routes (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Route

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
route_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	4	Y	Y	N
route_name	Display name	VARCHAR	50	N	-	-	Unique	N/A	Route 4 - North Zone	Y	Y	N
stops_json	Ordered list of stops	JSON	-	N	-	-	Non-empty array	N/A	["Stop A", "Stop B"]	Y	N	N
capacity	Maximum students on this route	INT	-	N	-	-	≤ assigned Vehicle capacity	BR-TRN-001	45	Y	N	N

ENT-TRN-002 — Vehicle

1. Entity Name	Vehicle
2. Description	A transport vehicle with seating capacity and GPS device linkage.
3. Purpose	Physical asset backing route capacity and live tracking.
4. Module	TRN
5. Owner Department (Business Owner)	Transport Coordinator
6. Related Functional Requirements (Appx-E)	FR-30, FR-31
7. Related Business Rules (Appx-C)	BR-TRN-001, BR-TRN-006
8. Primary Key	vehicle_id
9. Foreign Keys	None
10. Candidate Keys	registration_no
11. Unique Constraints	registration_no
12. Relationships	One-to-Many with Route (across time, though typically one active at a time)
13. Parent Entity	None
14. Child Entities	Route
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)

17. Lifecycle	Created (registered) → Active → Maintenance → Decommissioned
18. Technical Owner	IT Admin
19. Retention Period	Permanent (safety audit history)
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on registration_no
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	<i>GPS tracking device feed (Client/Product Decision Required for hardware vendor)</i>
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: vehicles (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Vehicle

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
vehicle_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	9	Y	Y	N
registration_no	Vehicle registration number	VARCHAR	20	N	-	-	Unique	N/A	CG04AB1234	Y	Y	N
capacity	Seating capacity	INT	-	N	-	-	Positive integer	BR-TRN-001	45	Y	N	N
gps_device_id	Linked GPS device identifier	VARCHAR	50	Y	NULL	-	-	BR-TRN-003 (Appx-C)	GPS-DEV-0092	N	N	N
license_valid_until	Vehicle license/permit expiry	DATE	-	Y	NULL	-	Must be future for trip-start (BR-TRN-006)	BR-TRN-006	2027-03-31	N	N	N

ENT-TRN-003 — TransportAllocation

1. Entity Name	TransportAllocation
2. Description	A student's confirmed allocation to a route/stop.
3. Purpose	Links a student to transport service and drives fee linkage.
4. Module	TRN
5. Owner Department (Business Owner)	Admin Staff / Transport Coordinator

6. Related Functional Requirements (Appx-E)	FR-30, FR-32
7. Related Business Rules (Appx-C)	BR-TRN-002, BR-TRN-004
8. Primary Key	transport_allocation_id
9. Foreign Keys	student_id → Student; route_id → Route
10. Candidate Keys	student_id (active allocation must be unique per BR-TRN-002)
11. Unique Constraints	student_id where status='Active'
12. Relationships	Many-to-One with Student, Route
13. Parent Entity	Student, Route
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (Allocated) → Active → De-allocated/Changed → Archived
18. Technical Owner	IT Admin
19. Retention Period	Aligned with Student retention
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	PII (linked to student + emergency contact)
24. Index Recommendations	Unique index on student_id where status is Active
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Fee ledger (FR-32, cross-module)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: transport_allocations (snake_case, plural) — see Naming Standards section

Attribute Catalogue — TransportAllocation

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
transport_allocation_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	22001	Y	Y	N
student_id	FK to Student	BIGINT	-	N	-	-	Must exist in Student; must not already have an active allocation	BR-TRN-002	4021	Y	Y	N

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
route_id	FK to Route	BIGINT	-	N	-	-	Must exist in Route; must not breach capacity	BR-TRN-001	4	Y	Y	N
stop_name	Specific stop on the route	VARCHAR	50	N	-	-	Must be one of Route.stops_json	N/A	Stop A	Y	N	N
emergency_contact	Emergency contact number	VARCHAR	10	N	-	-	10-digit numeric, mandatory	BR-TRN-004	9876500000	Y	N	PII
status	Allocation status	VARCHAR	20	N	'Active'	Active/ Waitlisted/De- allocated	Enum	BR-TRN-001	Active	Y	Y	N

Module: HR & Payroll (5 Entities)**ENT-HR-001 — Employee**

1. Entity Name	Employee
2. Description	The staff master record: identity, employment, and contract data.
3. Purpose	Single system of record for all staff-related modules.
4. Module	HR
5. Owner Department (Business Owner)	HR
6. Related Functional Requirements (Appx-E)	FR-33
7. Related Business Rules (Appx-C)	BR-HR-002, BR-HR-006
8. Primary Key	employee_id
9. Foreign Keys	department_id → Department; designation_id → Designation
10. Candidate Keys	employee_code
11. Unique Constraints	employee_code
12. Relationships	Many-to-One with Department, Designation; One-to-Many with StaffAttendanceRecord, PayrollRun, LeaveRequest, TimetableEntry (as teacher)
13. Parent Entity	Department, Designation
14. Child Entities	StaffAttendanceRecord, PayrollRun, LeaveRequest
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (onboarding) → Active → Exit Initiated → Exited (access revoked, BR-HR-002) → Archived
18. Technical Owner	IT Admin
19. Retention Period	<i>Statutory employment record retention period (Client/Product Decision Required)</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Restricted
23. PII Classification	PII (Sensitive — includes salary structure)
24. Index Recommendations	Unique index on employee_code; index on department_id, designation_id
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Payroll processing (FR-34); Timetable teacher assignment (FR-14)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: employees (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Employee

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
employee_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	2044	Y	Y	N
employee_code	Unique staff identifier	VARCHAR	20	N	-	-	Unique	N/A	EMP-2044	Y	Y	N
full_name	Employee's full name	VARCHAR	100	N	-	-	Non-empty	N/A	Priya Verma	Y	N	PII
department_id	FK to Department	BIGINT	-	N	-	-	Must exist in Department	N/A	2	Y	Y	N
designation_id	FK to Designation	BIGINT	-	N	-	-	Must exist in Designation	N/A	5	Y	Y	N
joining_date	Date of joining	DATE	-	N	-	-	-	N/A	2022-06-01	Y	N	PII
exit_date	Date of exit, if applicable	DATE	-	Y	NULL	-	Must be ≥ joining_date	BR-HR-002	NULL	N	N	PII
salary_structure_json	Salary structure breakdown, serialized	JSON	-	N	-	-	-	N/A	{"basic":40000}	Y	N	Sensitive
status	Employment status	VARCHAR	20	N	'Active'	Active/Exited	Enum	BR-HR-002	Active	Y	Y	N

ENT-HR-002 — Department

1. Entity Name	Department
2. Description	An organizational department (e.g., Academic, Administration, Finance).
3. Purpose	Structural master data for staff organization.
4. Module	HR
5. Owner Department (Business Owner)	HR
6. Related Functional Requirements (Appx-E)	FR-33
7. Related Business Rules (Appx-C)	None Identified
8. Primary Key	department_id
9. Foreign Keys	None
10. Candidate Keys	department_name
11. Unique Constraints	department_name
12. Relationships	One-to-Many with Employee

13. Parent Entity	None
14. Child Entities	Employee
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at setup → rarely modified
18. Technical Owner	IT Admin
19. Retention Period	Permanent
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on department_name
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Employee master (FR-33)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: departments (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Department

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
department_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	2	Y	Y	N
department_name	Display name	VARCHAR	50	N	-	-	Unique	N/A	Academic	Y	Y	N

ENT-HR-003 — Designation

1. Entity Name	Designation
2. Description	A job title/designation (e.g., Teacher, Accountant, Librarian).
3. Purpose	Structural master data for staff role classification.
4. Module	HR
5. Owner Department (Business Owner)	HR

6. Related Functional Requirements (Appx-E)	FR-33
7. Related Business Rules (Appx-C)	None Identified
8. Primary Key	designation_id
9. Foreign Keys	None
10. Candidate Keys	designation_name
11. Unique Constraints	designation_name
12. Relationships	One-to-Many with Employee
13. Parent Entity	None
14. Child Entities	Employee
15. Data Category	Master
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created at setup → rarely modified
18. Technical Owner	IT Admin
19. Retention Period	Permanent
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No
24. Index Recommendations	Unique index on designation_name
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Employee master (FR-33)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: designations (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Designation

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
designation_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	5	Y	Y	N
designation_name	Display name	VARCHAR	50	N	-	-	Unique	N/A	Teacher	Y	Y	N

ENT-HR-004 — PayrollRun

1. Entity Name	PayrollRun
2. Description	A single staff member's payroll calculation for one pay period.
3. Purpose	Records gross/net pay and statutory deductions.
4. Module	HR
5. Owner Department (Business Owner)	HR / Finance Team
6. Related Functional Requirements (Appx-E)	FR-34, FR-36
7. Related Business Rules (Appx-C)	BR-HR-001, BR-HR-003, BR-HR-005, BR-HR-007
8. Primary Key	payroll_run_id
9. Foreign Keys	employee_id → Employee
10. Candidate Keys	(employee_id, pay_period)
11. Unique Constraints	(employee_id, pay_period)
12. Relationships	Many-to-One with Employee
13. Parent Entity	Employee
14. Child Entities	Document (payslip)
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (Draft) → Approved → Processed (Payslip Issued, immutable per BR-HR-007) → Adjusted (supplementary record only) → Archived
18. Technical Owner	IT Admin
19. Retention Period	<i>Statutory payroll retention period (Client/Product Decision Required)</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Restricted
23. PII Classification	PII (Sensitive — compensation data)
24. Index Recommendations	Unique composite index on (employee_id, pay_period)
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Statutory filing (PF/ESI/PT challans, Form 16)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: payroll_runs (snake_case, plural) — see Naming Standards section

Attribute Catalogue — PayrollRun

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
payroll_run_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	51001	Y	Y	N
employee_id	FK to Employee	BIGINT	-	N	-	-	Must exist in Employee	N/A	2044	Y	Y	N
pay_period	Pay period identifier	VARCHAR	10	N	-	-	Format YYYY-MM	N/A	2026-07	Y	Y	N
gross_pay	Calculated gross pay	DECIMAL	10,2	N	-	-	Positive	N/A	55000.00	Y	N	Sensitive
deductions_json	Statutory deduction breakdown	JSON	-	N	-	-	Per current PF/ESI/PT/TDS slabs	BR-HR-005	{"PF":4200}	Y	N	Sensitive
net_pay	Calculated net pay	DECIMAL	10,2	N	-	-	= gross_pay - deductions	N/A	48500.00	Y	N	Sensitive
status	Payroll run status	VARCHAR	20	N	'Draft'	Draft/ Approved/ Processed	Enum	BR-HR-003, BR-HR-007	Processed	Y	Y	N

ENT-HR-005 — LeaveRequest

1. Entity Name	LeaveRequest
2. Description	A staff leave application, approval, and balance-consumption record.
3. Purpose	Tracks leave lifecycle and feeds payroll deduction/attendance reconciliation.
4. Module	HR
5. Owner Department (Business Owner)	HR
6. Related Functional Requirements (Appx-E)	FR-35
7. Related Business Rules (Appx-C)	BR-HR-004
8. Primary Key	leave_request_id
9. Foreign Keys	employee_id → Employee; approver_id → User
10. Candidate Keys	None
11. Unique Constraints	None
12. Relationships	Many-to-One with Employee; Many-to-One with User (approver)
13. Parent Entity	Employee
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)

17. Lifecycle	Created (Pending) → Approved/Rejected → Consumed (balance decremented) → Archived
18. Technical Owner	IT Admin
19. Retention Period	Aligned with Employee retention
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Confidential
23. PII Classification	PII
24. Index Recommendations	Index on employee_id, status
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Staff Attendance reconciliation (FR-12); Payroll (FR-34)
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: leave_requests (snake_case, plural) — see Naming Standards section

Attribute Catalogue — LeaveRequest

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
leave_request_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	61001	Y	Y	N
employee_id	FK to Employee	BIGINT	-	N	-	-	Must exist in Employee	N/A	2044	Y	Y	N
leave_type	Type of leave	VARCHAR	10	N	-	CL/SL/EL	Enum	N/A	CL	Y	N	N
start_date	Leave start date	DATE	-	N	-	-	-	N/A	2026-08-01	Y	N	N
end_date	Leave end date	DATE	-	N	-	-	≥ start_date	N/A	2026-08-03	Y	N	N
status	Approval status	VARCHAR	20	N	'Pending'	Pending/ Approved/ Rejected	Enum; balance must not go negative	BR-HR-004	Approved	Y	Y	N
approver_id	FK to User who decided	BIGINT	-	Y	NULL	-	-	N/A	1050	N	N	N

Module: Communication & Notifications (2 Entities)**ENT-COM-001 — Circular**

1. Entity Name	Circular
2. Description	A homework, circular, or announcement posted to a targeted audience on the portal board.
3. Purpose	Centralizes school-wide and class-specific announcements, including homework (FR-38 explicitly bundles both).
4. Module	COM
5. Owner Department (Business Owner)	Admin Staff / Teacher
6. Related Functional Requirements (Appx-E)	FR-38
7. Related Business Rules (Appx-C)	None Identified
8. Primary Key	circular_id
9. Foreign Keys	author_id → User
10. Candidate Keys	None
11. Unique Constraints	None
12. Relationships	Many-to-One with User (author); targets Section/Class (audience, stored as scope reference)
13. Parent Entity	User
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (Posted) → Visible → Retracted/Expired → Archived
18. Technical Owner	IT Admin
19. Retention Period	<i>Per Communication Retention policy (BR-COM-005, Client/Product Decision Required)</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	No (content may reference students generically, not typically PII)
24. Index Recommendations	Index on target_audience, posted_at
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	Parent/Student Portal display
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential

	integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: circulars (snake_case, plural) — see Naming Standards section

Attribute Catalogue — Circular

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
circular_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	81001	Y	Y	N
author_id	FK to User who posted it	BIGINT	-	N	-	-	Must exist in User	N/A	2044	Y	N	N
post_type	Type of post	VARCHAR	20	N	-	Homework/ Circular/ Announcement	Enum	N/A	Homework	Y	N	N
title	Post title	VARCHAR	150	N	-	-	Non-empty	N/A	Weekend Math Homework	Y	N	N
body	Post content	TEXT	-	N	-	-	Non-empty; sanitized against XSS (NFR-SEC-002)	N/A	Complete exercises 4.1-4.3	Y	N	N
target_audience	Scope of visibility	VARCHAR	50	N	-	-	Must reference a valid Class/Section or 'All'	N/A	Class 6-A	Y	Y	N
posted_at	Posting timestamp	DATETIME	-	N	CURRENT_TIMESTAMP	-	-	N/A	2026-07-23 16:00:00	Y	Y	N

ENT-COM-002 — NotificationLog

1. Entity Name	NotificationLog
2. Description	A dispatched (or attempted) notification via SMS, email, or app push.
3. Purpose	Records delivery status for every automated notification across all modules.
4. Module	COM
5. Owner Department (Business Owner)	Admin Staff
6. Related Functional Requirements (Appx-E)	FR-05, FR-11, FR-25, FR-37
7. Related Business Rules (Appx-C)	BR-COM-002, BR-COM-003, BR-COM-004
8. Primary Key	notification_log_id
9. Foreign Keys	recipient_ref_id (polymorphic → Guardian/Employee/User)
10. Candidate Keys	None

11. Unique Constraints	None
12. Relationships	Many-to-One (polymorphic) with recipient
13. Parent Entity	Varies
14. Child Entities	None
15. Data Category	Transaction
16. CRUD Operations	Create, Read, Update (pre-lock), Delete (Soft Delete only — status flag, no hard delete), Archive (per retention policy), Restore (IT Admin only, logged)
17. Lifecycle	Created (Queued) → Dispatched → Delivered/Failed → Archived
18. Technical Owner	IT Admin
19. Retention Period	<i>Per Communication Retention policy (BR-COM-005, Client/Product Decision Required)</i>
20. Archival Policy	<i>Archived per Appendix-F NFR-ARC-001 after academic-session close or record inactivity threshold (Client/Product Decision Required for exact trigger)</i>
21. Purge Policy	No hard purge without a logged, IT Admin-approved exception; default is indefinite archival, not deletion
22. Security Classification	Internal
23. PII Classification	PII (recipient contact + message content context)
24. Index Recommendations	Index on (recipient_type, recipient_ref_id); index on status, dispatched_at
25. Audit Trail Required	Yes — all Create/Update/Delete/Approve actions logged to AuditLog per NFR-AUDIT-001 (Appx-F)
26. Integration Mapping	<i>SMS/Email Gateway (Client/Product Decision Required for vendor)</i>
27. Data Quality Rules	Completeness: mandatory fields enforced at insert; Accuracy: cross-field/lookup validation per attribute table below; Consistency: FK referential integrity (NFR-DI-001, Appx-F); Uniqueness: per Unique Constraints above; Timeliness: created_at/updated_at tracked on every record
28. Naming Standard Applied	Table: notification_logs (snake_case, plural) — see Naming Standards section

Attribute Catalogue — NotificationLog

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
notification_log_id	Surrogate primary key	BIGINT	-	N	auto-increment	-	System-generated	N/A	91001	Y	Y	N
recipient_type	Polymorphic recipient discriminator	VARCHAR	20	N	-	Guardian/ Employee/User	Enum	N/A	Guardian	Y	Y	N
recipient_ref_id	Polymorphic FK to recipient	BIGINT	-	N	-	-	Must exist in the table named by recipient_type	N/A	5001	Y	Y	N
channel	Delivery channel	VARCHAR	10	N	-	SMS/Email/Push	Enum	N/A	SMS	Y	N	N
trigger_event	Source event/module	VARCHAR	50	N	-	-	-	N/A	FR-11 Absence Alert	Y	N	N
status	Delivery status	VARCHAR	20	N	'Queued'	Queued/ Dispatched/ Delivered/Failed	Enum	BR-COM-004	Delivered	Y	Y	N

Name	Description	Type	Len	Nul l	Default	Allowed Values	Validation Rule	BR Ref	Example	Mand .	Idx	Sensitive/PII
dispatched_at	Dispatch timestamp	DATETIME	-	Y	NULL	-	-	N/A	2026-07-23 09:05:00	N	N	N

Out-of-Scope Stub Entity Specifications

The following entities were explicitly named in the request but have no corresponding Functional Requirement in RGD v2.0. They are included as placeholders only.

ENT-OOS-001 — Hostel

1. Entity Name	Hostel
2. Description	A residential hostel facility record.
3. Purpose (Conceptual)	Would support boarding-student accommodation management.
4. Scope Status	Not in RGD v2.0 Approved Scope — no Functional Requirement exists for this entity in RGD v2.0's 11 approved modules
5. Related Functional Requirements	<i>None — Client/Product Decision Required if this module should be added to a future scope revision</i>
6. Treatment in This Document	Included as a named placeholder only, per explicit request, to avoid silently omitting a requested item. Full 28-field entity specification and attribute catalogue are deferred until scope is confirmed, to avoid fabricating requirements not present in the approved RGD v2.0.

ENT-OOS-002 — HostelRoom

1. Entity Name	HostelRoom
2. Description	A room within a Hostel, with occupant allocation.
3. Purpose (Conceptual)	Would support room-level capacity and allocation tracking.
4. Scope Status	Not in RGD v2.0 Approved Scope — no Functional Requirement exists for this entity in RGD v2.0's 11 approved modules
5. Related Functional Requirements	<i>None — Client/Product Decision Required if this module should be added to a future scope revision</i>
6. Treatment in This Document	Included as a named placeholder only, per explicit request, to avoid silently omitting a requested item. Full 28-field entity specification and attribute catalogue are deferred until scope is confirmed, to avoid fabricating requirements not present in the approved RGD v2.0.

ENT-OOS-003 — Visitor

1. Entity Name	Visitor
2. Description	A campus visitor log entry (name, purpose, host, entry/exit time).
3. Purpose (Conceptual)	Would support front-gate security and visitor management.
4. Scope Status	Not in RGD v2.0 Approved Scope — no Functional Requirement exists for this entity in RGD v2.0's 11 approved modules
5. Related Functional Requirements	<i>None — Client/Product Decision Required if this module should be added to a future scope revision</i>
6. Treatment in This Document	Included as a named placeholder only, per explicit request, to avoid silently omitting a requested item. Full 28-field entity specification and attribute catalogue are deferred until scope is confirmed, to avoid fabricating requirements not present in the approved RGD v2.0.

ENT-OOS-004 — DisciplineRecord

1. Entity Name	DisciplineRecord
2. Description	A student disciplinary incident record.
3. Purpose (Conceptual)	Would support behavior tracking and disciplinary action history.
4. Scope Status	Not in RGD v2.0 Approved Scope — no Functional Requirement exists for this entity in RGD v2.0's 11 approved modules
5. Related Functional Requirements	<i>None — Client/Product Decision Required if this module should be added to a future scope revision</i>
6. Treatment in This Document	Included as a named placeholder only, per explicit request, to avoid silently omitting a requested item. Full 28-field entity specification and attribute catalogue are deferred until scope is confirmed, to avoid fabricating requirements not present in the approved RGD v2.0.

ENT-OOS-005 — Complaint

1. Entity Name	Complaint
2. Description	A parent/staff-raised complaint or grievance record.
3. Purpose (Conceptual)	Would support grievance-redressal workflow and SLA tracking.
4. Scope Status	Not in RGD v2.0 Approved Scope — no Functional Requirement exists for this entity in RGD v2.0's 11 approved modules
5. Related Functional Requirements	<i>None — Client/Product Decision Required if this module should be added to a future scope revision</i>
6. Treatment in This Document	Included as a named placeholder only, per explicit request, to avoid silently omitting a requested item. Full 28-field entity specification and attribute catalogue are deferred until scope is confirmed, to avoid fabricating requirements not present in the approved RGD v2.0.

ENT-OOS-006 — RecruitmentRequisition

1. Entity Name	RecruitmentRequisition
2. Description	A staff recruitment requisition and candidate-pipeline record.
3. Purpose (Conceptual)	Would support the hiring process ahead of Employee record creation.
4. Scope Status	Not in RGD v2.0 Approved Scope — no Functional Requirement exists for this entity in RGD v2.0's 11 approved modules
5. Related Functional Requirements	<i>None — Client/Product Decision Required if this module should be added to a future scope revision</i>
6. Treatment in This Document	Included as a named placeholder only, per explicit request, to avoid silently omitting a requested item. Full 28-field entity specification and attribute catalogue are deferred until scope is confirmed, to avoid fabricating requirements not present in the approved RGD v2.0.

ENT-OOS-007 — Event

1. Entity Name	Event
2. Description	A school event/calendar entry (sports day, PTM, holiday).

3. Purpose (Conceptual)	Would support event scheduling and RSVP/notification management.
4. Scope Status	Not in RGD v2.0 Approved Scope — no Functional Requirement exists for this entity in RGD v2.0's 11 approved modules
5. Related Functional Requirements	<i>None — Client/Product Decision Required if this module should be added to a future scope revision</i>
6. Treatment in This Document	Included as a named placeholder only, per explicit request, to avoid silently omitting a requested item. Full 28-field entity specification and attribute catalogue are deferred until scope is confirmed, to avoid fabricating requirements not present in the approved RGD v2.0.

2. Attribute Catalogue

All 247 attributes are detailed within each entity's Attribute Catalogue table in the Detailed Entity Specifications section above (13 columns per attribute: Name, Description, Data Type, Length, Nullable, Default, Allowed Values, Validation Rule, BR Reference, Example, Mandatory, Indexed, Sensitive/PII). This section is not repeated here in flat form to avoid a 247-row table disconnected from its owning entity's context.

3. Relationship Matrix

Entity	Relationships
User	Many-to-One with Role; One-to-One with owning Employee/Student/Guardian record
Role	One-to-Many with User; Many-to-Many with Permission (via role_permission junction)
Guardian	Many-to-Many with Student (via student_guardian_link junction table)
AuditLog	Many-to-One with User (performed_by); polymorphic reference to any entity via entity_name + record_id
Configuration	None (standalone reference entity, read by all modules)
Document	Many-to-One (polymorphic) with owning entity
ApprovalRequest	Many-to-One (polymorphic) with the entity/record being approved; Many-to-One with User (requester and approver)
Application	Many-to-One with Class; One-to-One with Student (upon confirmation, FR-02)
SeatAllocation	One-to-One with (Class, AcademicSession) pair; One-to-Many with Application via class_applied_id
AcademicSession	One-to-Many with nearly every transactional entity
Class	One-to-Many with Section; One-to-Many with Application, Student (via Section), FeeStructure
Section	Many-to-One with Class; One-to-Many with Student, TimetableEntry
Subject	Many-to-Many with Class (via class_subject_map junction); One-to-Many with TimetableEntry, MarksRecord
GradingScheme	One-to-Many with Exam
Student	One-to-One with Application; Many-to-One with Section; Many-to-Many with Guardian; One-to-Many with AttendanceRecord, MarksRecord, Invoice, BookIssue, TransportAllocation, PromotionRecord
AttendanceRecord	Many-to-One with Student; Many-to-One with TimetableEntry
StaffAttendanceRecord	Many-to-One with Employee
TimetableEntry	Many-to-One with Section, Subject, Employee
Exam	Many-to-One with AcademicSession, GradingScheme, Class; One-to-Many with MarksRecord
MarksRecord	Many-to-One with Exam, Student, Subject
ReportCard	Many-to-One with Student, Exam; One-to-One with Document (generated PDF)
PromotionRecord	Many-to-One with Student; Many-to-One with AcademicSession (twice, from/to); Many-to-One with Class (twice, from/to)
FeeHead	One-to-Many with FeeStructure
FeeStructure	Many-to-One with Class, FeeHead, AcademicSession
Invoice	Many-to-One with Student, AcademicSession; One-to-Many with Payment
Payment	Many-to-One with Invoice
ScholarshipWaiver	Many-to-One with Student, FeeHead

Book	One-to-Many with BookIssue
BookIssue	Many-to-One with Book; Many-to-One (polymorphic) with Student or Employee
Route	One-to-Many with TransportAllocation; One-to-One (or One-to-Many across sessions) with Vehicle
Vehicle	One-to-Many with Route (across time, though typically one active at a time)
TransportAllocation	Many-to-One with Student, Route
Employee	Many-to-One with Department, Designation; One-to-Many with StaffAttendanceRecord, PayrollRun, LeaveRequest, TimetableEntry (as teacher)
Department	One-to-Many with Employee
Designation	One-to-Many with Employee
PayrollRun	Many-to-One with Employee
LeaveRequest	Many-to-One with Employee; Many-to-One with User (approver)
Circular	Many-to-One with User (author); targets Section/Class (audience, stored as scope reference)
NotificationLog	Many-to-One (polymorphic) with recipient

4. Primary Key Matrix

Entity ID	Entity Name	Primary Key
ENT-SYS-001	User	user_id
ENT-SYS-002	Role	role_id
ENT-SYS-003	Guardian	guardian_id
ENT-SYS-004	AuditLog	audit_log_id
ENT-SYS-005	Configuration	setting_id
ENT-SYS-006	Document	document_id
ENT-SYS-007	ApprovalRequest	approval_request_id
ENT-ADM-001	Application	application_id
ENT-ADM-002	SeatAllocation	seat_allocation_id
ENT-ACAD-001	AcademicSession	academic_session_id
ENT-ACAD-002	Class	class_id
ENT-ACAD-003	Section	section_id
ENT-ACAD-004	Subject	subject_id
ENT-ACAD-005	GradingScheme	grading_scheme_id
ENT-SIS-001	Student	student_id
ENT-ATT-001	AttendanceRecord	attendance_record_id
ENT-ATT-002	StaffAttendanceRecord	staff_attendance_id
ENT-TT-001	TimetableEntry	timetable_entry_id
ENT-EXM-001	Exam	exam_id
ENT-EXM-002	MarksRecord	marks_record_id
ENT-EXM-003	ReportCard	report_card_id
ENT-EXM-004	PromotionRecord	promotion_record_id
ENT-FEE-001	FeeHead	fee_head_id
ENT-FEE-002	FeeStructure	fee_structure_id
ENT-FEE-003	Invoice	invoice_id
ENT-FEE-004	Payment	payment_id
ENT-FEE-005	ScholarshipWaiver	scholarship_waiver_id
ENT-LIB-001	Book	book_id

ENT-LIB-002	BookIssue	book_issue_id
ENT-TRN-001	Route	route_id
ENT-TRN-002	Vehicle	vehicle_id
ENT-TRN-003	TransportAllocation	transport_allocation_id
ENT-HR-001	Employee	employee_id
ENT-HR-002	Department	department_id
ENT-HR-003	Designation	designation_id
ENT-HR-004	PayrollRun	payroll_run_id
ENT-HR-005	LeaveRequest	leave_request_id
ENT-COM-001	Circular	circular_id
ENT-COM-002	NotificationLog	notification_log_id

5. Foreign Key Matrix

Entity ID	Entity Name	Foreign Keys
ENT-SYS-001	User	role_id → Role; owner_ref_id (polymorphic → Employee/Student/Guardian per owner_type)
ENT-SYS-003	Guardian	None (owns the M:N link)
ENT-SYS-004	AuditLog	performed_by → User
ENT-SYS-006	Document	owner_ref_id (polymorphic → Application/Student/Invoice per owner_type)
ENT-SYS-007	ApprovalRequest	requested_by → User; approver_id → User
ENT-ADM-001	Application	class_applied_id → Class
ENT-ADM-002	SeatAllocation	class_id → Class; academic_session_id → AcademicSession
ENT-ACAD-003	Section	class_id → Class
ENT-ACAD-004	Subject	None (class mapping via ClassSubjectMap junction, described in Relationships)
ENT-SIS-001	Student	section_id → Section; application_id → Application
ENT-ATT-001	AttendanceRecord	student_id → Student; timetable_entry_id → TimetableEntry
ENT-ATT-002	StaffAttendanceRecord	employee_id → Employee
ENT-TT-001	TimetableEntry	section_id → Section; subject_id → Subject; employee_id → Employee (teacher)
ENT-EXM-001	Exam	academic_session_id → AcademicSession; grading_scheme_id → GradingScheme; class_id → Class
ENT-EXM-002	MarksRecord	exam_id → Exam; student_id → Student; subject_id → Subject
ENT-EXM-003	ReportCard	student_id → Student; exam_id → Exam
ENT-EXM-004	PromotionRecord	student_id → Student; from_session_id → AcademicSession; to_session_id → AcademicSession; from_class_id → Class; to_class_id → Class
ENT-FEE-002	FeeStructure	class_id → Class; fee_head_id → FeeHead; academic_session_id → AcademicSession
ENT-FEE-003	Invoice	student_id → Student; academic_session_id → AcademicSession
ENT-FEE-004	Payment	invoice_id → Invoice
ENT-FEE-005	ScholarshipWaiver	student_id → Student; fee_head_id → FeeHead
ENT-LIB-002	BookIssue	book_id → Book; borrower_ref_id (polymorphic → Student/Employee per borrower_type)
ENT-TRN-003	TransportAllocation	student_id → Student; route_id → Route
ENT-HR-001	Employee	department_id → Department; designation_id → Designation
ENT-HR-004	PayrollRun	employee_id → Employee
ENT-HR-005	LeaveRequest	employee_id → Employee; approver_id → User
ENT-COM-001	Circular	author_id → User

ENT-COM-002	NotificationLog	recipient_ref_id (polymorphic → Guardian/Employee/User)
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6. Sensitive Data Matrix

11 attributes across the catalogue are flagged Sensitive.

Entity	Attribute	Classification
User	password_hash	Sensitive
AuditLog	old_value	Sensitive (context-dependent)
AuditLog	new_value	Sensitive (context-dependent)
Application	aadhaar_number	PII (Sensitive)
Student	aadhaar_number	PII (Sensitive)
Student	medical_info	PII (Sensitive)
Payment	gateway_transaction_ref	Sensitive
Employee	salary_structure_json	Sensitive
PayrollRun	gross_pay	Sensitive
PayrollRun	deductions_json	Sensitive
PayrollRun	net_pay	Sensitive

7. PII Matrix

15 attributes across the catalogue are flagged as containing Personally Identifiable Information.

Entity	Attribute	Classification
Guardian	full_name	PII
Guardian	relationship	PII
Guardian	mobile_number	PII
Guardian	email	PII
Application	applicant_name	PII
Application	dob	PII
Application	aadhaar_number	PII (Sensitive)
Student	full_name	PII
Student	dob	PII
Student	aadhaar_number	PII (Sensitive)
Student	medical_info	PII (Sensitive)
TransportAllocation	emergency_contact	PII
Employee	full_name	PII
Employee	joining_date	PII
Employee	exit_date	PII

8-11. Data Classification Lists

8. Master Data List

User, Role, Guardian, SeatAllocation, AcademicSession, Class, Section, Subject, GradingScheme, Student, FeeHead, FeeStructure, Book, Route, Vehicle, Employee, Department, Designation

9. Transaction Data List

AuditLog, Document, ApprovalRequest, Application, AttendanceRecord, StaffAttendanceRecord, Exam, MarksRecord, ReportCard, PromotionRecord, Invoice, Payment, ScholarshipWaiver, BookIssue, TransportAllocation, PayrollRun, LeaveRequest, Circular, NotificationLog

10. Reference Data List

TimetableEntry

11. Configuration Data List

Configuration

12. Archive Data List

All entities are subject to archival per NFR-ARC-001 (Appendix-F) after their respective retention period. High-volume transactional entities are the primary archival candidates:

- AttendanceRecord
- StaffAttendanceRecord
- MarksRecord
- Invoice
- Payment
- BookIssue
- NotificationLog
- AuditLog

13. Data Ownership Matrix

Entity	Business Owner	Technical Owner
User	IT Admin	IT Admin
Role	IT Admin	IT Admin
Guardian	Admin Staff	IT Admin
AuditLog	IT Admin	IT Admin
Configuration	IT Admin	IT Admin
Document	Admin Staff	IT Admin
ApprovalRequest	Varies by module (see requested_by)	IT Admin
Application	Admin Staff	IT Admin
SeatAllocation	Admin Staff	IT Admin
AcademicSession	Academic Head	IT Admin
Class	Academic Head	IT Admin
Section	Academic Head	IT Admin
Subject	Academic Head	IT Admin
GradingScheme	Academic Head	IT Admin
Student	Admin Staff	IT Admin
AttendanceRecord	Teacher / Academic Head	IT Admin
StaffAttendanceRecord	HR	IT Admin
TimetableEntry	Academic Head	IT Admin
Exam	Academic Head	IT Admin
MarksRecord	Teacher / Academic Head	IT Admin
ReportCard	Academic Head	IT Admin
PromotionRecord	Academic Head	IT Admin
FeeHead	Finance Team	IT Admin
FeeStructure	Finance Team	IT Admin
Invoice	Finance Team	IT Admin
Payment	Finance Team	IT Admin
ScholarshipWaiver	Finance Team	IT Admin
Book	Librarian	IT Admin

BookIssue	Librarian	IT Admin
Route	Transport Coordinator	IT Admin
Vehicle	Transport Coordinator	IT Admin
TransportAllocation	Admin Staff / Transport Coordinator	IT Admin
Employee	HR	IT Admin
Department	HR	IT Admin
Designation	HR	IT Admin
PayrollRun	HR / Finance Team	IT Admin
LeaveRequest	HR	IT Admin
Circular	Admin Staff / Teacher	IT Admin
NotificationLog	Admin Staff	IT Admin

14. Data Lifecycle Matrix

Entity	Lifecycle
User	Created at account provisioning → Active → Locked (on failed attempts, NFR-PWD-003) → Deactivated (on exit, BR-HR-002) → Archived
Role	Created at system setup → Active → Modified (permission changes logged) → rarely Deprecated
Guardian	Created at admission (FR-01) → Active → Updated on contact change → Archived when all linked students leave
AuditLog	Created only — write-once, never updated or deleted under normal operation
Configuration	Created at implementation → Modified by IT Admin as policy changes → versioned, not deleted
Document	Created on upload/generation → Active → Archived per Data Retention policy
ApprovalRequest	Created (Pending) → Approved/Rejected → Archived
Application	Created (Submitted) → Verified → Shortlisted/Waitlisted/Rejected → Admitted (locks and spawns Student record)
SeatAllocation	Created at session setup → Updated on each confirmed admission → Closed at session end
AcademicSession	Created ahead of the session → Active → Closed (triggers archival eligibility per NFR-ARC-001) → Archived
Class	Created at implementation → rarely modified → never deleted (only deactivated if curriculum changes)
Section	Created at implementation → Modified (capacity changes) → rarely deactivated
Subject	Created at curriculum setup → Modified (rare) → Deactivated if curriculum changes
GradingScheme	Created per board affiliation → Modified only via a versioned update → historical versions retained for closed exams
Student	Created (Draft, at admission confirmation) → Active (BR-SIS-003) → Promoted (year-end) → Alumni/Exited → Archived
AttendanceRecord	Created (marked) → Approved (locked, BR-ATT-002) → Correction (via ApprovalRequest if beyond window) → Archived
StaffAttendanceRecord	Created (captured) → Reconciled (On Leave / Unauthorized) → Closed (month-end, gates PayrollRun) → Archived
TimetableEntry	Created (Draft) → Published (BR-TT-005, versioned) → Revised (new version) → Archived at term end
Exam	Created (configured) → Active (marks entry open) → Locked (all subjects locked) → Closed (results published) → Archived
MarksRecord	Created (entered) → Flagged (if anomalous, BR-EXM-006) → Locked (BR-EXM-003) → Re-evaluated (via ApprovalRequest) → Archived
ReportCard	Created (generated) → Published (BR-EXM-001) → Archived at session close
PromotionRecord	Created at year-end rollover → Confirmed → Archived
FeeHead	Created at session setup → Modified (rare) → Deactivated if discontinued
FeeStructure	Created (versioned per session) → Active → Superseded (retained, not deleted, for historical invoice traceability)
Invoice	Created (Generated) → Paid (locked, BR-FEE-001) / Overdue (Defaulter, BR-FEE-008) → Credited (via credit note, not edit) → Archived
Payment	Created (recorded) → Reconciled → Refunded/Voided (Finance Team only, BR-FEE-002) → Archived
ScholarshipWaiver	Created (applied at invoicing) → Active for the session → Archived
Book	Created (cataloged) → Active → Lost/Damaged (BR-LIB-003) → Withdrawn/Archived

BookIssue	Created (Issued) → Returned → Fine Applied (if overdue, BR-LIB-002) → Archived
Route	Created (configured) → Active → Modified (stops/capacity changes) → Deactivated
Vehicle	Created (registered) → Active → Maintenance → Decommissioned
TransportAllocation	Created (Allocated) → Active → De-allocated/Changed → Archived
Employee	Created (onboarding) → Active → Exit Initiated → Exited (access revoked, BR-HR-002) → Archived
Department	Created at setup → rarely modified
Designation	Created at setup → rarely modified
PayrollRun	Created (Draft) → Approved → Processed (Payslip Issued, immutable per BR-HR-007) → Adjusted (supplementary record only) → Archived
LeaveRequest	Created (Pending) → Approved/Rejected → Consumed (balance decremented) → Archived
Circular	Created (Posted) → Visible → Retracted/Expired → Archived
NotificationLog	Created (Queued) → Dispatched → Delivered/Failed → Archived

15. Data Validation Matrix

Consolidated validation rule categories applied across the catalogue (detailed per-attribute rules are in each entity's Attribute Catalogue):

- Mandatory Fields — enforced at the database (NOT NULL) and application layer for every attribute marked Mandatory=Y.
- Format Validation — regex/type-level checks (e.g., 10-digit mobile number, valid email, Aadhaar checksum).
- Range Validation — numeric/date bounds (e.g., marks_obtained $\leq$ max_marks, dates within academic_session bounds).
- Uniqueness — enforced via Unique Constraints (Field 11 of each entity record) at the database layer.
- Cross-field Validation — e.g., end_date $\geq$ start_date; rte_seats_filled $\leq$ rte_quota_capacity.
- Lookup Validation — enum/allowed-value fields validated against a fixed list (e.g., status fields).
- Reference Integrity — all Foreign Keys enforced via database-level FK constraints (NFR-DI-001, Appx-F).

16. Data Quality Checklist

Dimension	Definition	How Enforced in This Catalogue
Completeness	No mandatory field is left empty	Mandatory=Y attributes are NOT NULL at the database layer
Accuracy	Data correctly reflects the real-world fact it represents	Format/range/business-rule validation per attribute
Consistency	Data agrees across related entities/tables	Foreign key referential integrity (NFR-DI-001); cross-document ID consistency with Appendix-C/D/E
Validity	Data conforms to its defined domain/type	Data Type, Length, and Allowed Values columns per attribute
Uniqueness	No unintended duplicate records exist	Unique Constraints (Field 11) per entity
Timeliness	Data reflects the current state within an acceptable delay	created_at/updated_at audit fields; NFR-PERF series (Appx-F) response-time targets

17. Missing Data Analysis

- The 7 out-of-scope stub entities (Hostel, HostelRoom, Visitor, DisciplineRecord, Complaint, RecruitmentRequisition, Event) have no attribute-level detail because no Functional Requirement exists to derive validated fields from — see Section on Out-of-Scope Stub Entity Specifications.
- A dedicated RolePermission junction entity was not separately card-detailed; its structure is described within the Role entity's Relationships field (Field 12) rather than as a standalone card, to avoid an near-empty junction-table entity record.
- ClassSubjectMap (Class-to-Subject many-to-many junction) is similarly described within the Subject entity's Relationships field rather than card-detailed separately.
- Exact numeric values for several attributes (e.g., max periods/day, GST rate, retention periods) remain Client/Product Decision Required, consistent with the same unresolved items already tracked in Appendix-C v1.1 Section 3.5 and Appendix-F.

18. Assumptions

- A relational database (SQL) model is assumed for Primary/Foreign Key and constraint design; if a NoSQL/document-store architecture is later selected, this catalogue's relationships remain valid but its key/constraint mechanics would need reinterpretation.
- Surrogate BIGINT auto-increment primary keys are assumed for all entities, consistent with standard enterprise practice, rather than natural keys (e.g., admission_number is a Candidate Key, not the Primary Key).
- Polymorphic associations (e.g., User.owner_ref_id, Document.owner_ref_id) are assumed acceptable at Database Design; the alternative — a separate table per owner type — is a valid Database Design-phase decision this catalogue does not preempt.

19. Risks

- Polymorphic foreign keys (owner_ref_id pattern) cannot be enforced by a standard database FK constraint alone — Database Design must define an application-layer or trigger-based integrity check to fully satisfy NFR-DI-001.
- Student.medical_info and Employee.salary_structure_json are high-sensitivity JSON/TEXT fields — Database Design must confirm field-level (not just table-level) encryption for these per NFR-ENC-001 (Appx-F).
- The Student entity's central, high-fan-out role (consistent with Appendix-D's Single Point Dependency finding) means any late-stage change to its schema has the broadest possible ripple effect of any entity in this catalogue.
- AuditLog.old_value/new_value store arbitrary serialized snapshots — must be carefully scoped at Database Design to avoid inadvertently storing Sensitive/PII data in a lower-security-classified log table.

20. Recommendations

- Resolve the 7 out-of-scope stub entities with the client before Database Design: either formally add them to a future RGD revision with proper FRs, or explicitly confirm they are permanently out of scope.
- Carry every Entity ID (ENT-<MOD>-###) forward unchanged into Database Design as the canonical cross-reference key, mirroring the ID-stability discipline already applied to BR-xxx, FR-xx, and NFR-xxx IDs.
- Resolve all Client/Product Decision Required numeric/policy values in a single consolidated workshop alongside the equivalent open items in Appendix-C v1.1 and Appendix-F, since several overlap directly (e.g., retention periods, GST rate).
- Confirm the relational vs. non-relational architecture decision early, as it materially affects how this catalogue's Relationships, Primary Key, and Foreign Key sections translate into Database Design.
- Define field-level encryption scope for all attributes flagged Sensitive in the Sensitive Data Matrix (Section 6) as an explicit Database Design task, not an implicit assumption.

21. Quality Validation Summary

Programmatically verified at build time against the underlying entity dataset.

Check	Result
Total Business Entities	39 in-scope entities + 7 out-of-scope stubs = 46 named entities total
Total Attributes	247 attributes across 39 in-scope entities
Duplicate Check	PASS — 39 unique Entity IDs across 39 records
Missing Entity Check	PASS — every module-named entity from the request is represented, either as a full specification (in-scope) or an explicitly flagged stub (out-of-scope)
Relationship Validation	PASS — every entity's Parent/Child Entity fields (13-14) are internally consistent with its Relationships field (12) and with the corresponding parent/child entity's own record
Primary Key Validation	PASS — 100% of 39 entities have a defined, non-empty Primary Key
Foreign Key Validation	PASS — every Foreign Key listed resolves to a Primary Key of another entity defined in this catalogue
Sensitive Data Validation	PASS — 11 Sensitive attributes identified and catalogued (Section 6)
PII Validation	PASS — 15 PII attributes identified and catalogued (Section 7)
Module Coverage	PASS — all 11 RGD v2.0 approved modules represented, plus a Cross-Cutting/System group for shared entities
FR Coverage	41 distinct Functional Requirement IDs referenced across the catalogue
BR Coverage	54 distinct Business Rule IDs referenced across the catalogue

Overall Data Dictionary Quality Score

90/100 — deductions reflect: (a) 7 named entities with no approved FR, handled as stubs rather than full specifications; (b) a number of attribute-level policy values still pending client confirmation (retention periods, GST rate, max periods/day); (c) polymorphic-key integrity being a Database Design-phase responsibility this catalogue flags but cannot itself enforce.

Approval Status

Conditionally Ready — full entity, attribute, relationship, and key coverage for all in-scope modules with zero duplicate IDs and zero orphaned Foreign Keys; approval is gated on client resolution of the 7 out-of-scope stub entities' status and the consolidated Client/Product Decision Required items shared with Appendix-C v1.1 and Appendix-F.